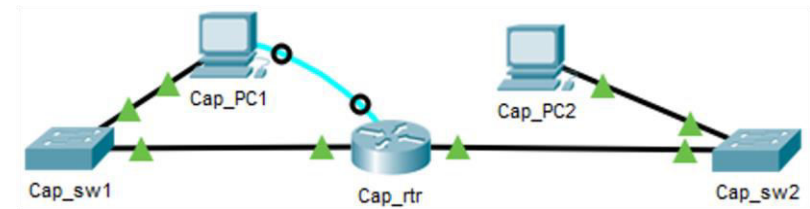
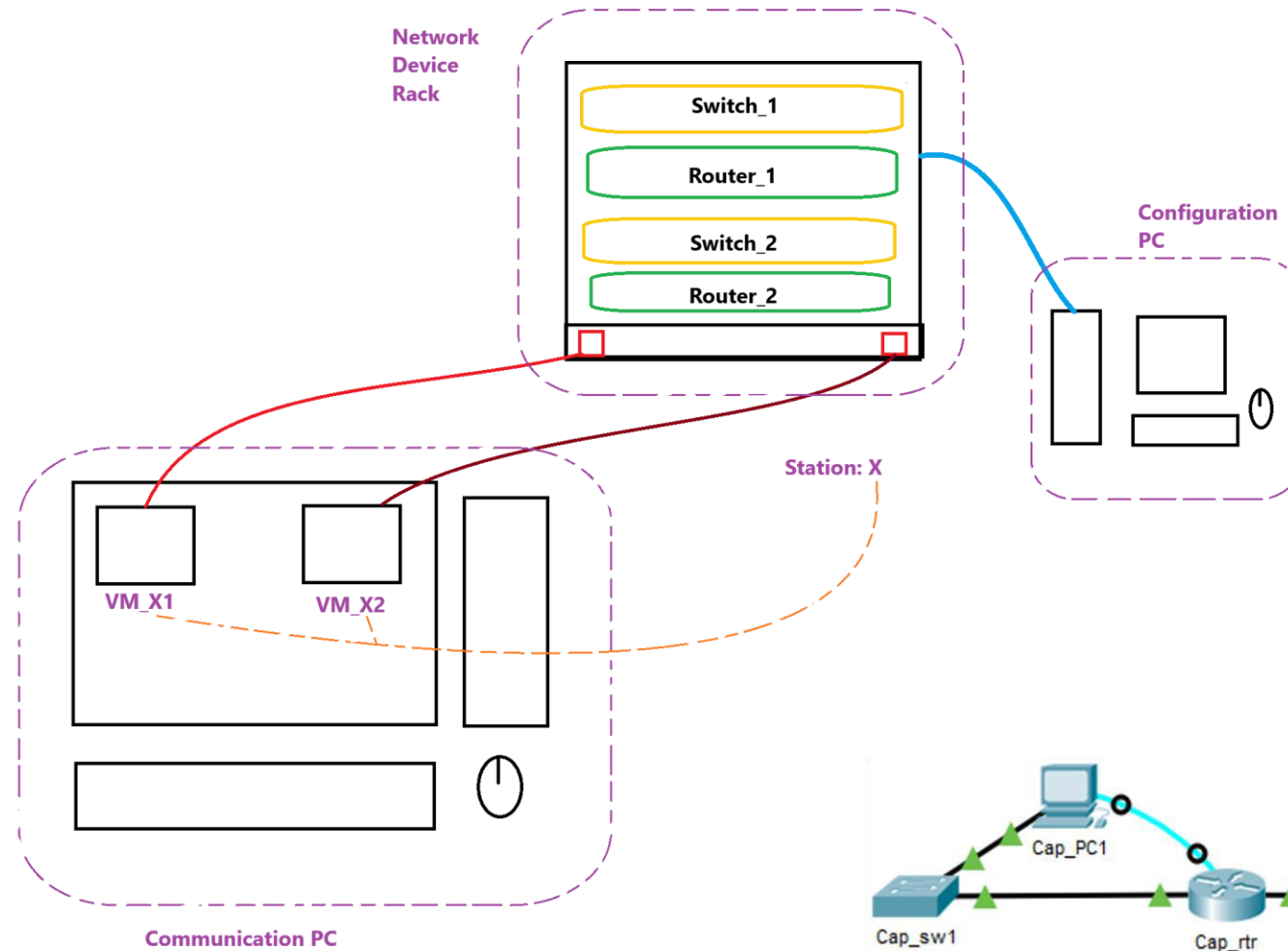


Network Lab

Setup and student guide

Lab setup (Diagram)

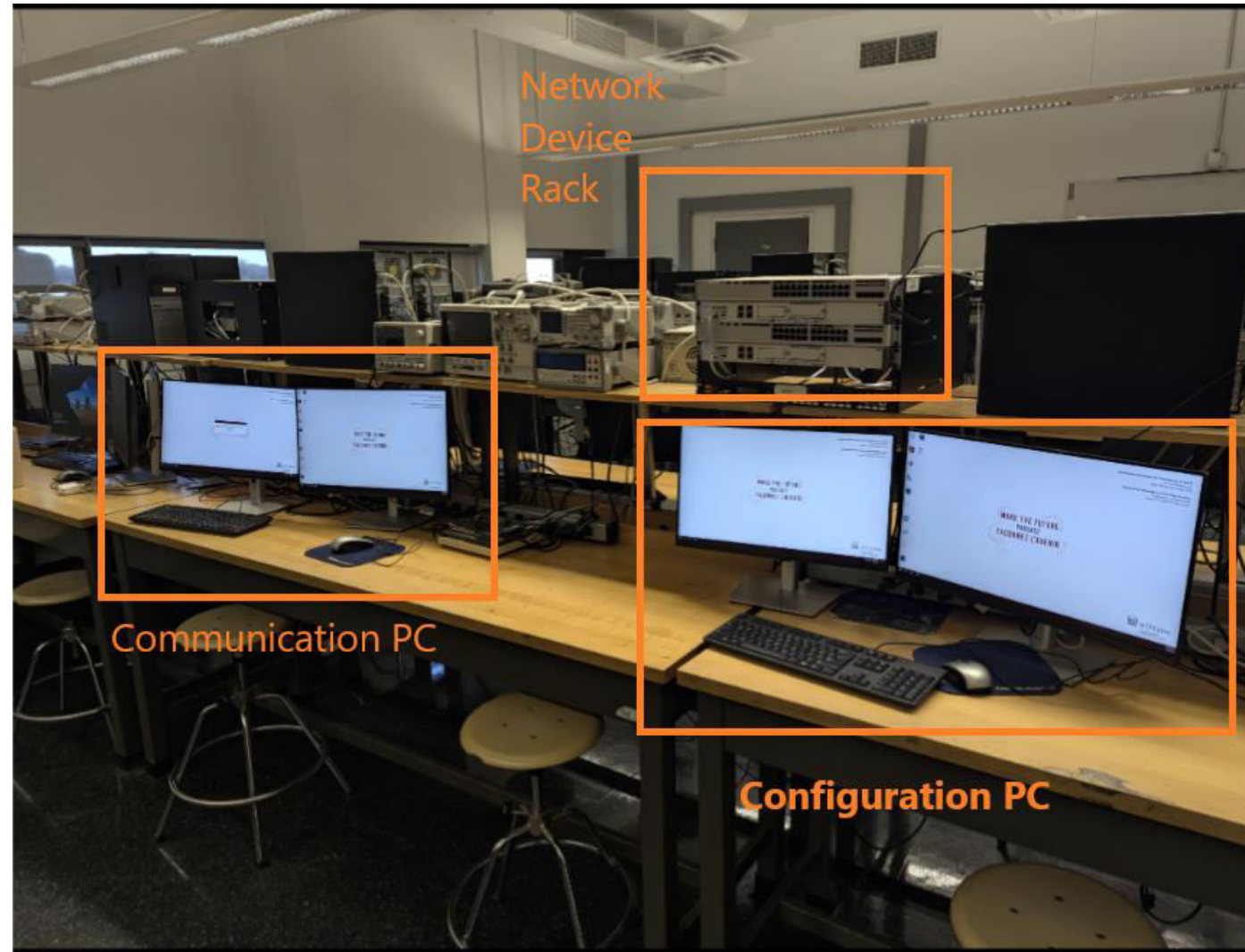


Lab infrastructure

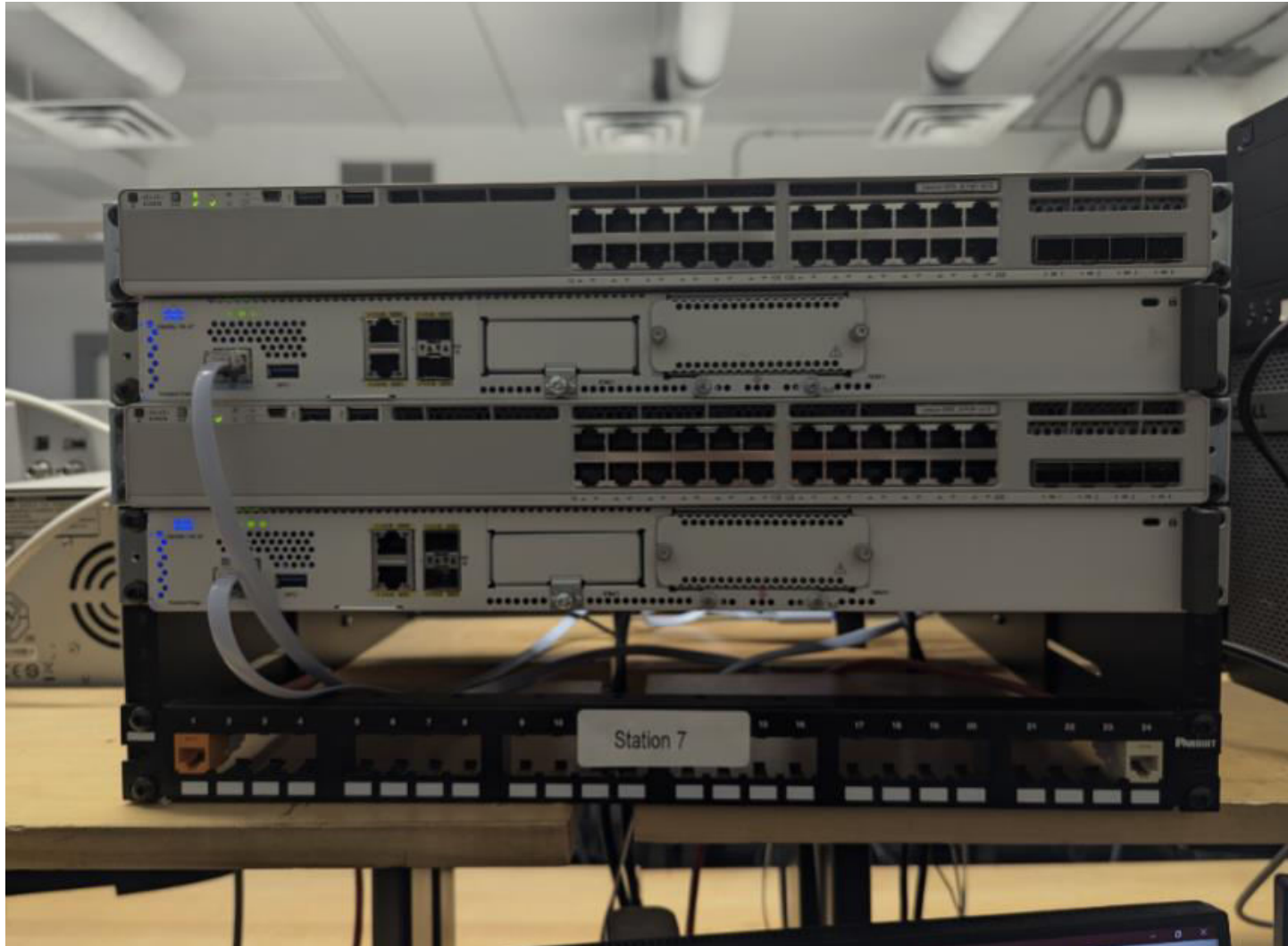
Station (Configuration PC)		VM1 (Communication PC)	VM2 (communication PC)
1	ID	11	12
	RDP IP	10.137.89.11	10.137.89.12
	Interface IP	192.168.11.11	192.168.12.12
2	ID	21	22
	RDP IP	10.137.89.21	10.137.89.22
	Interface IP	192.168.21.21	192.168.22.22
3	ID	31	32
	RDP IP	10.137.89.31	10.137.89.32
	Interface IP	192.168.31.31	192.168.32.32
4	ID	41	42
	RDP IP	10.137.89.41	10.137.89.42
	Interface IP	192.168.41.41	192.168.42.42
5	ID	51	52
	RDP IP	10.137.89.51	10.137.89.52
	Interface IP	192.168.51.51	192.168.52.52

Station (Configuration PC)		VM1 (Communication PC)	VM2 (Communication PC)
6	ID	61	62
	RDP IP	10.137.89.61	10.137.89.62
	Interface IP	192.168.61.61	192.168.62.62
7	ID	71	72
	RDP IP	10.137.89.71	10.137.89.72
	Interface IP	192.168.71.71	192.168.72.72
8	ID	81	82
	RDP IP	10.137.89.81	10.137.89.82
	Interface IP	192.168.81.81	192.168.82.82
9	ID	91	92
	RDP IP	10.137.89.91	10.137.89.92
	Interface IP	192.168.91.91	192.168.92.92
10	ID	101	102
	RDP IP	10.137.89.101	10.137.89.102
	Interface IP	192.168.101.101	192.168.102.102

Lab setup (real picture)



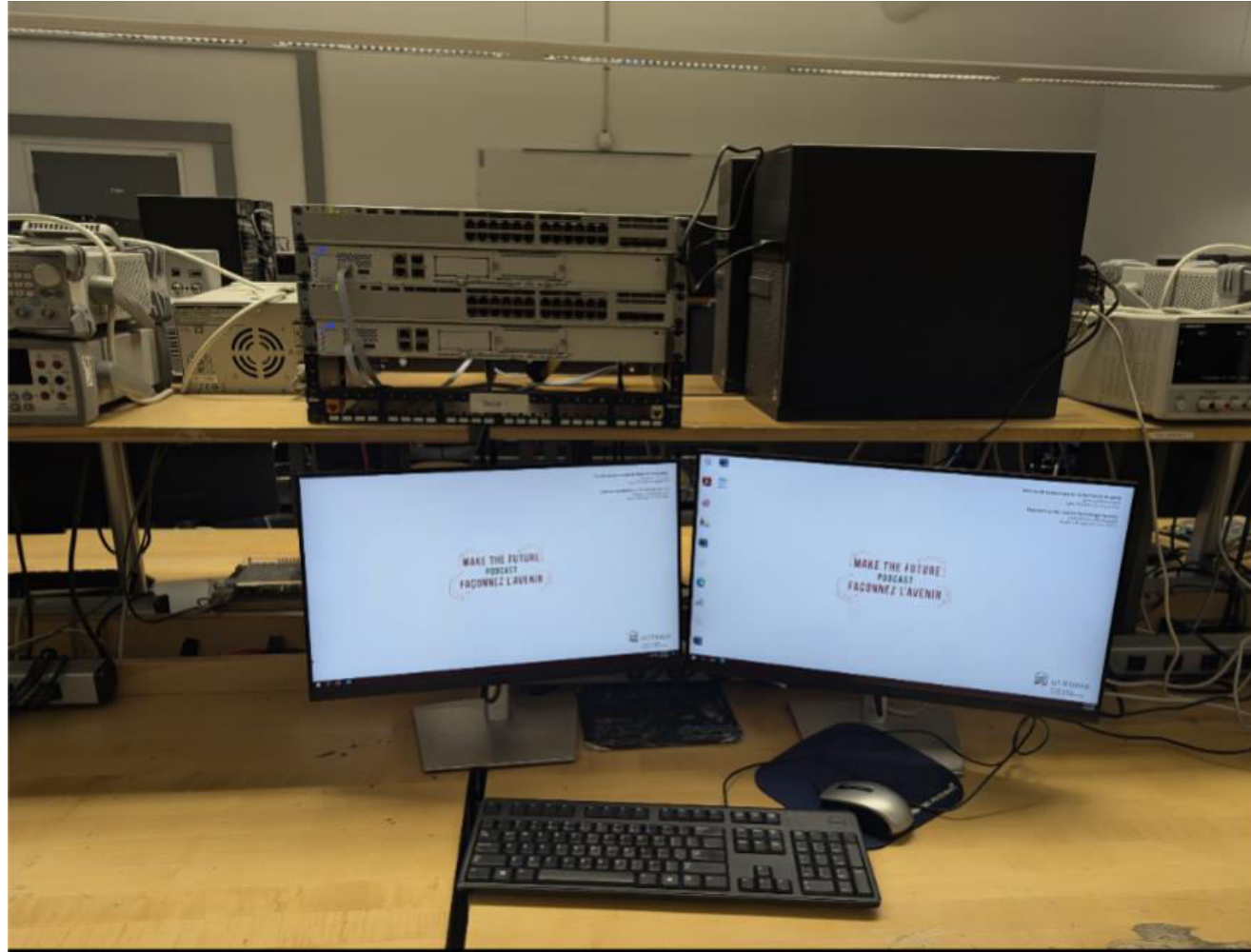
Device Rack (Station: X)



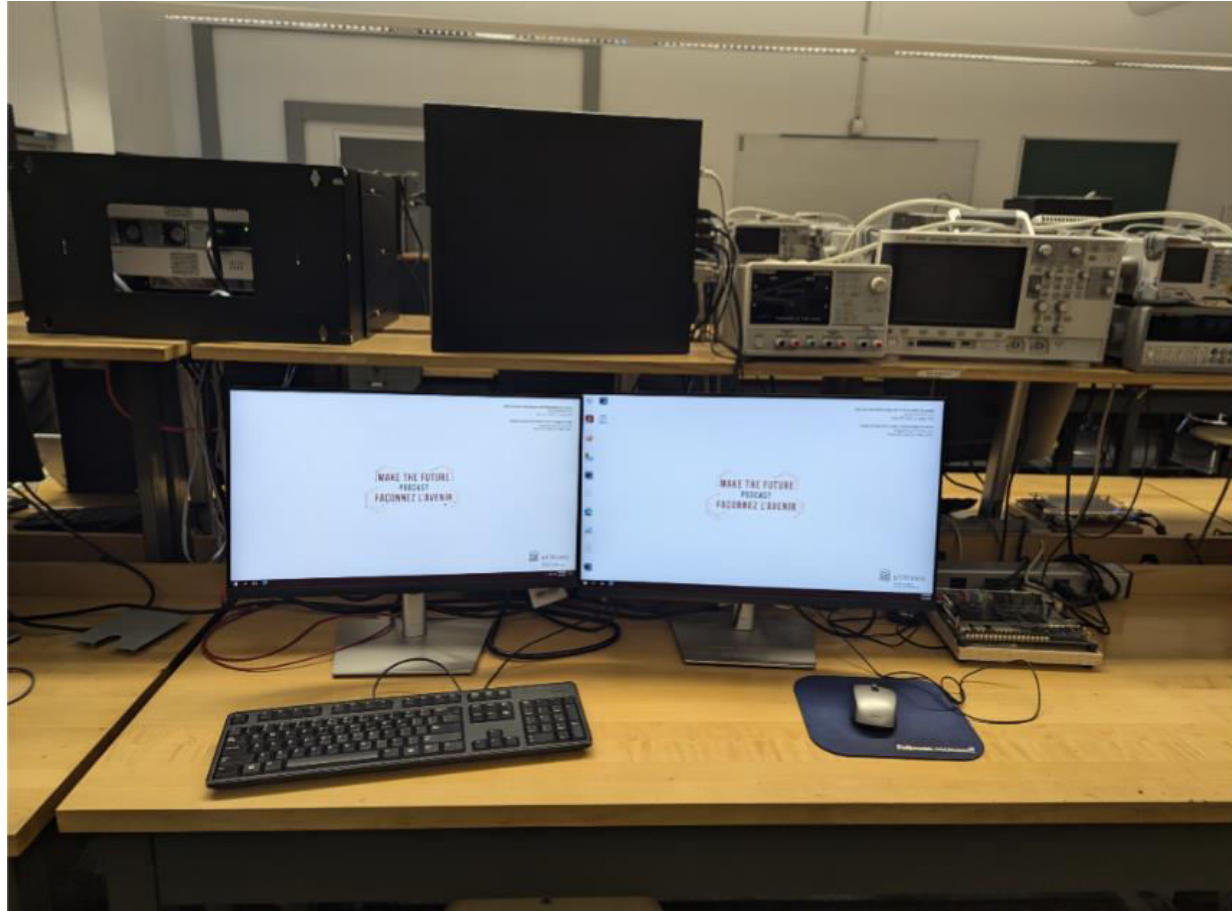
Device configuration PC

Note:

Network Device rack is connected to **Configuration PC** through a USB cable.



Communication VM-PC (VM1, VM2)



VM1

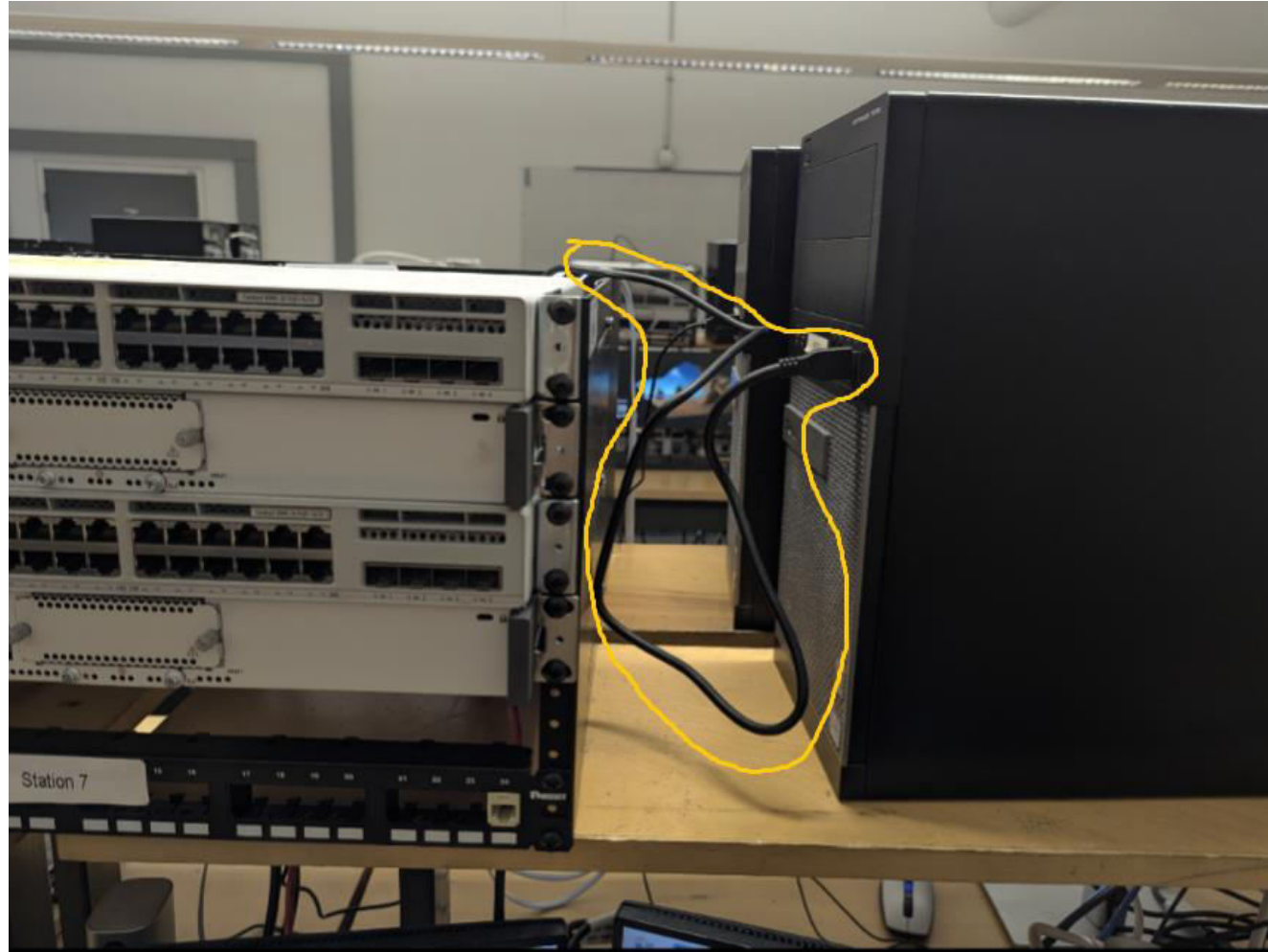
For station 7, it will be:
VM71

VM2

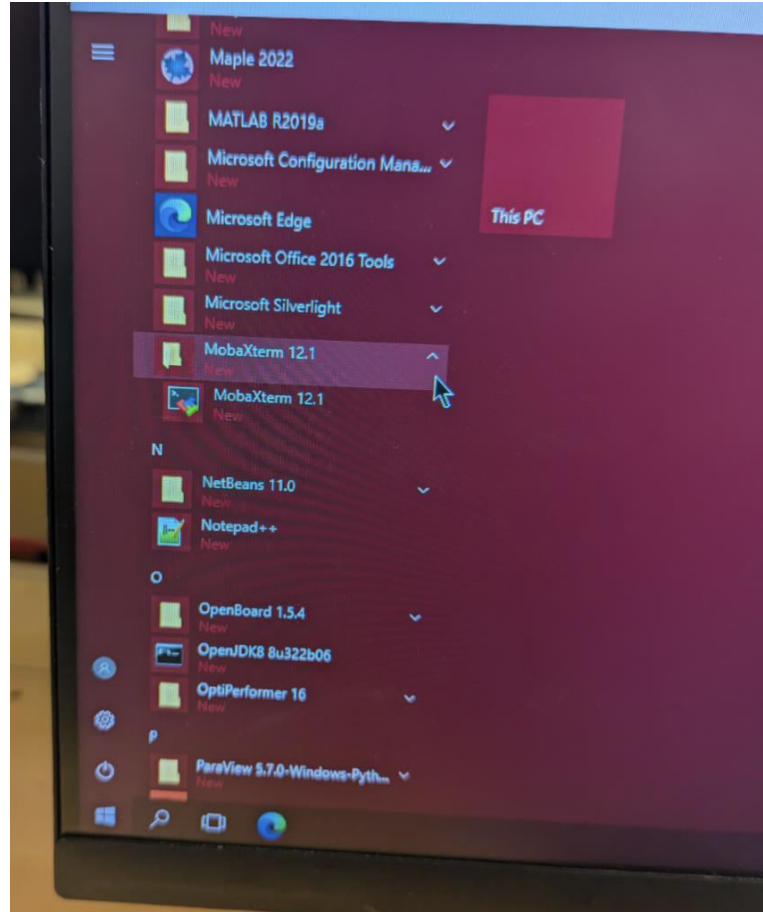
For station 7, it will be:
VM72

Device Rack connection

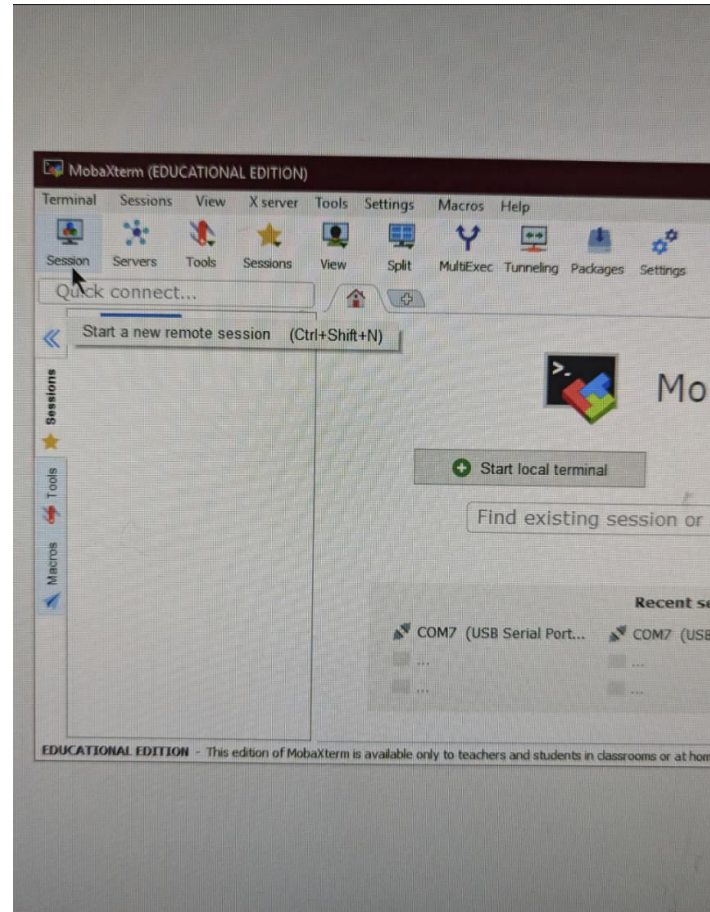
Make sure that the USB cable is connected.



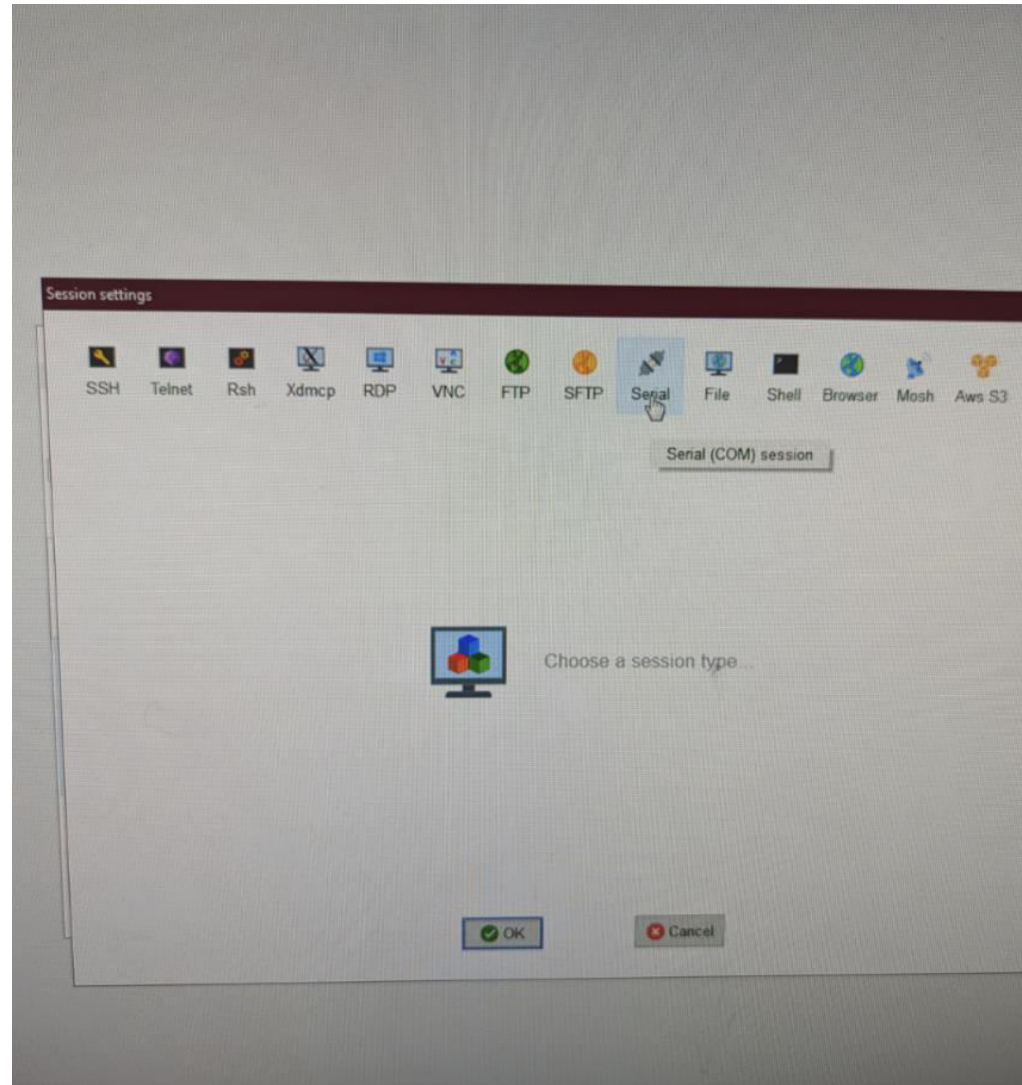
Configuration PC - setup



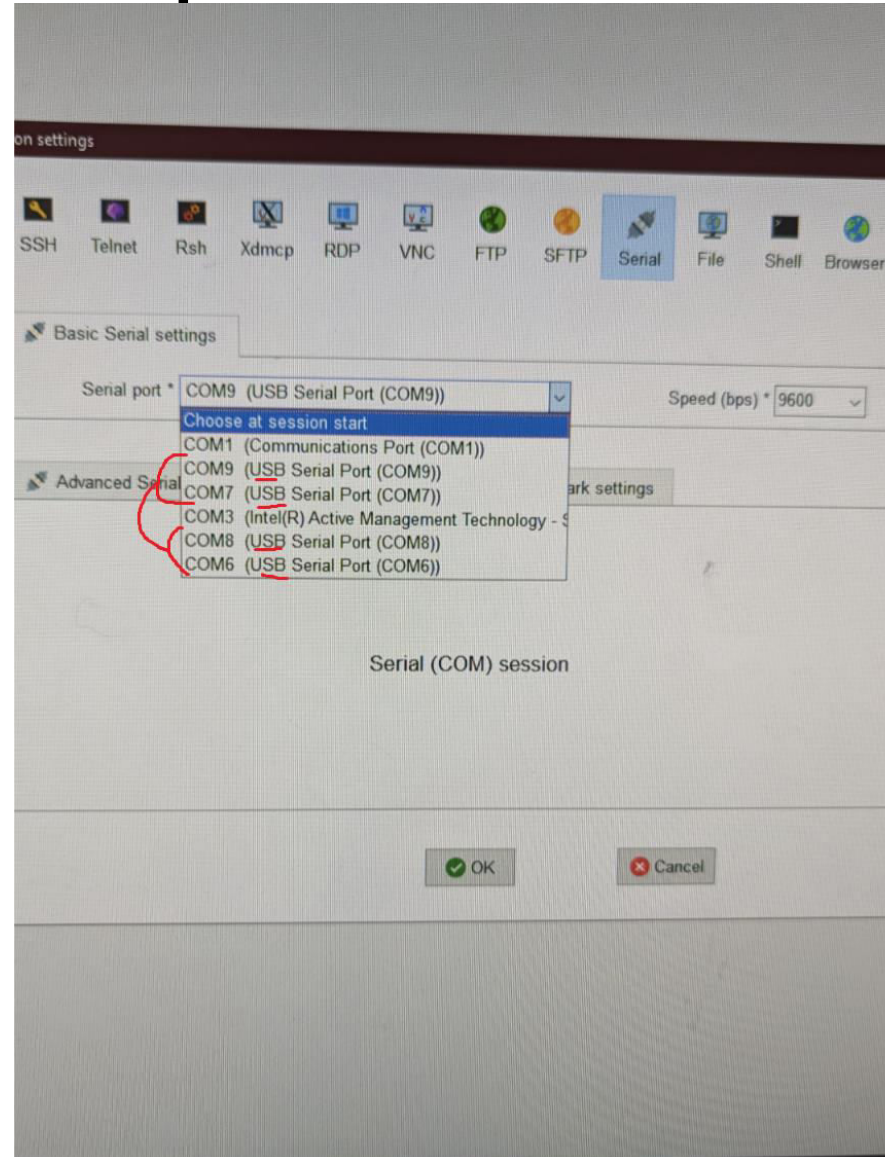
MobaXterm Setup



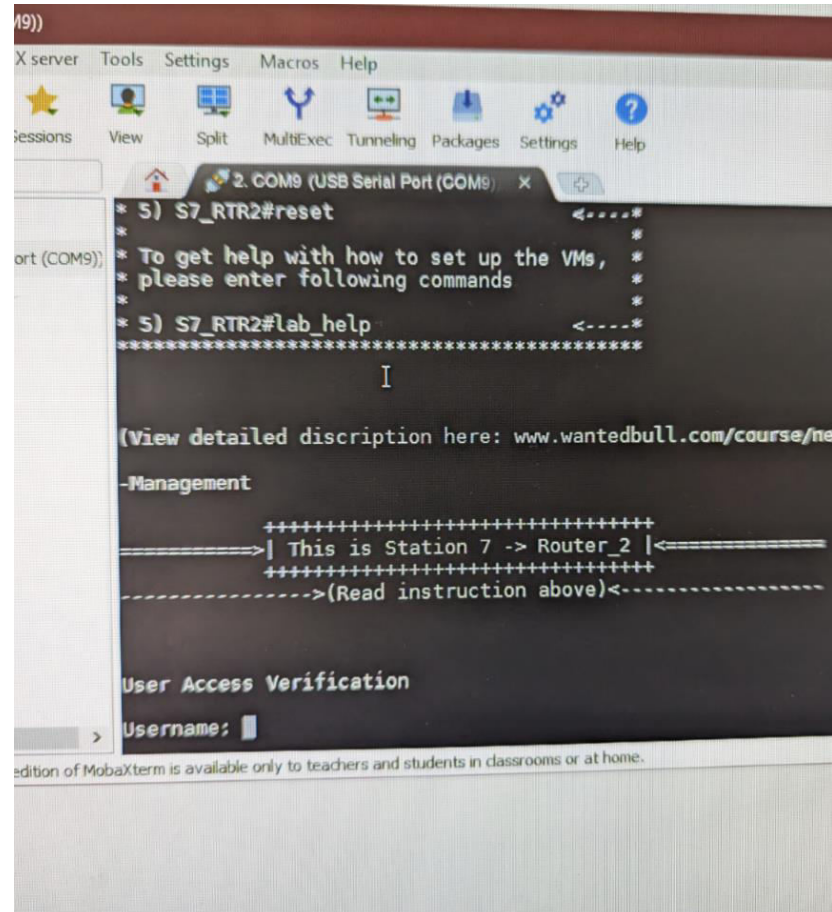
MobaXterm Setup



MobaXterm Setup



MobaXterm Setup



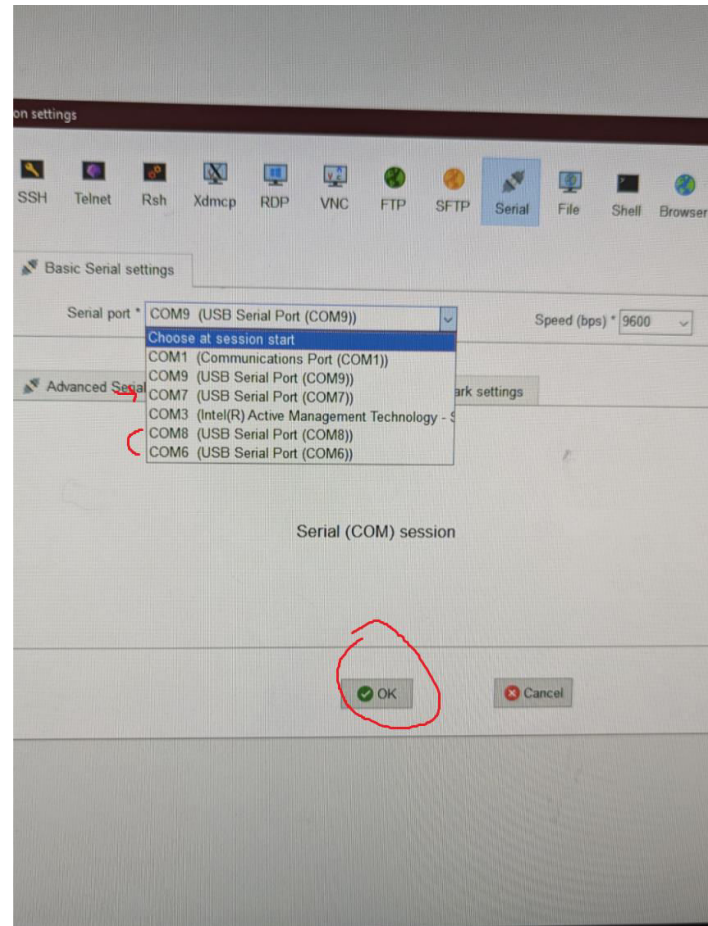
The screenshot shows the MobaXterm application window with a terminal session. The terminal displays a script for configuring a network device, likely a router. The script includes commands to reset the device and to display help information. The help information includes a link to a website for detailed instructions and a section for user access verification.

```
19))
X server  Tools  Settings  Macros  Help
Sessions  View  Split  MultiExec  Tunneling  Packages  Settings  Help

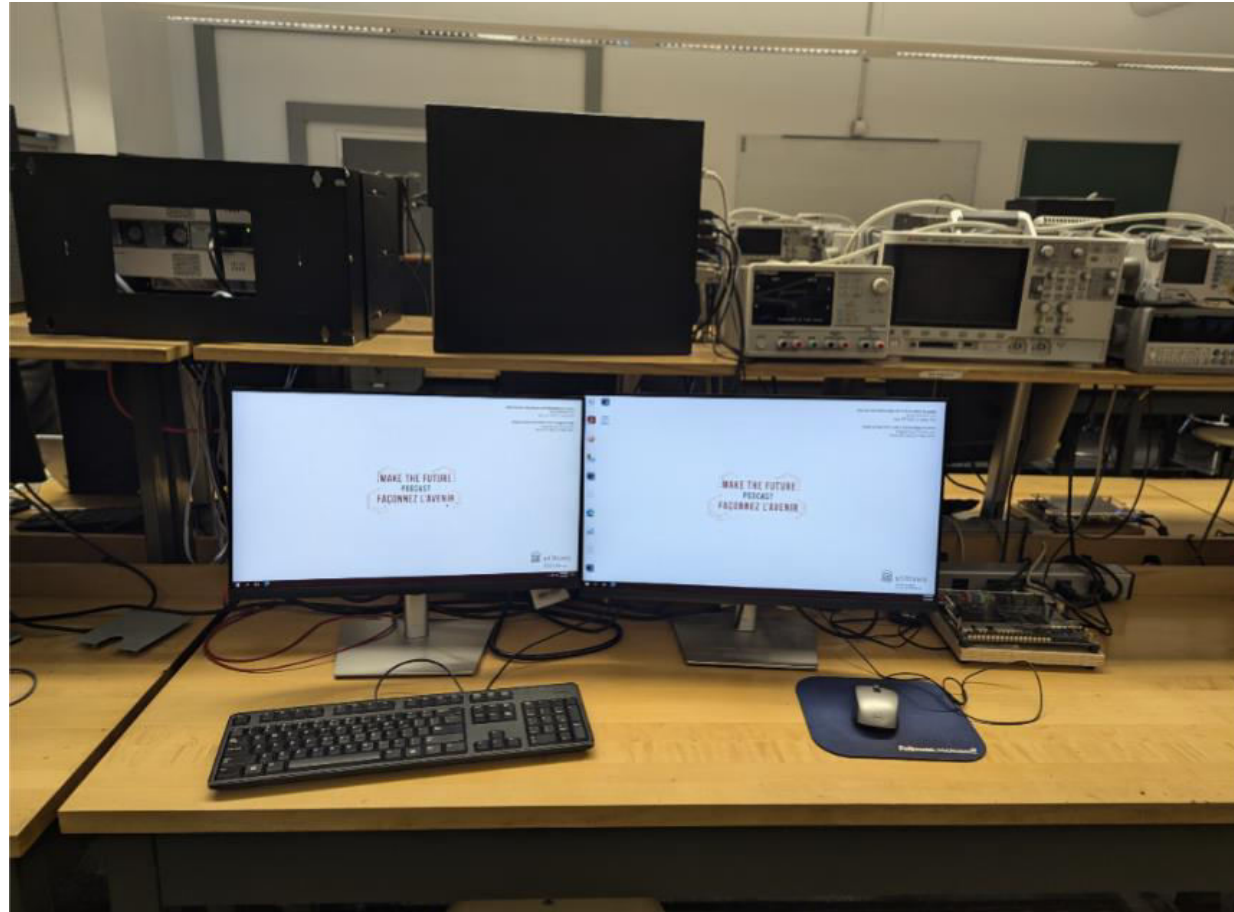
2. COM9 (USB Serial Port (COM9)) X
* S7_RTR2#reset
*
* To get help with how to set up the VMs,
* please enter following commands
*
* S7_RTR2#lab_help
*****
I
(View detailed discription here: www.wantedbull.com/course/ne)
-Management
+++++
=====>| This is Station 7 -> Router_2 |<=====
+++++
----->(Read instruction above)<-----
User Access Verification
> Username: 
```

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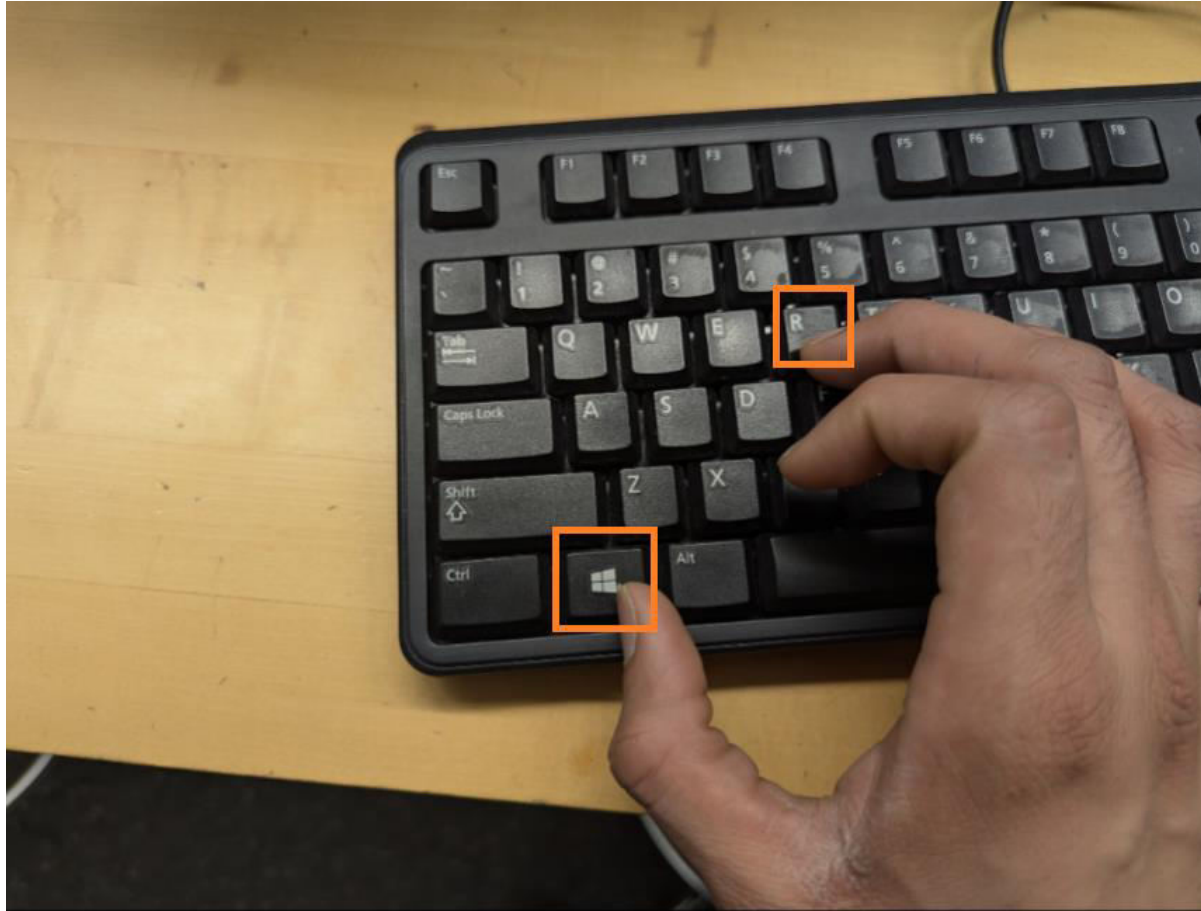
MobaXterm Setup



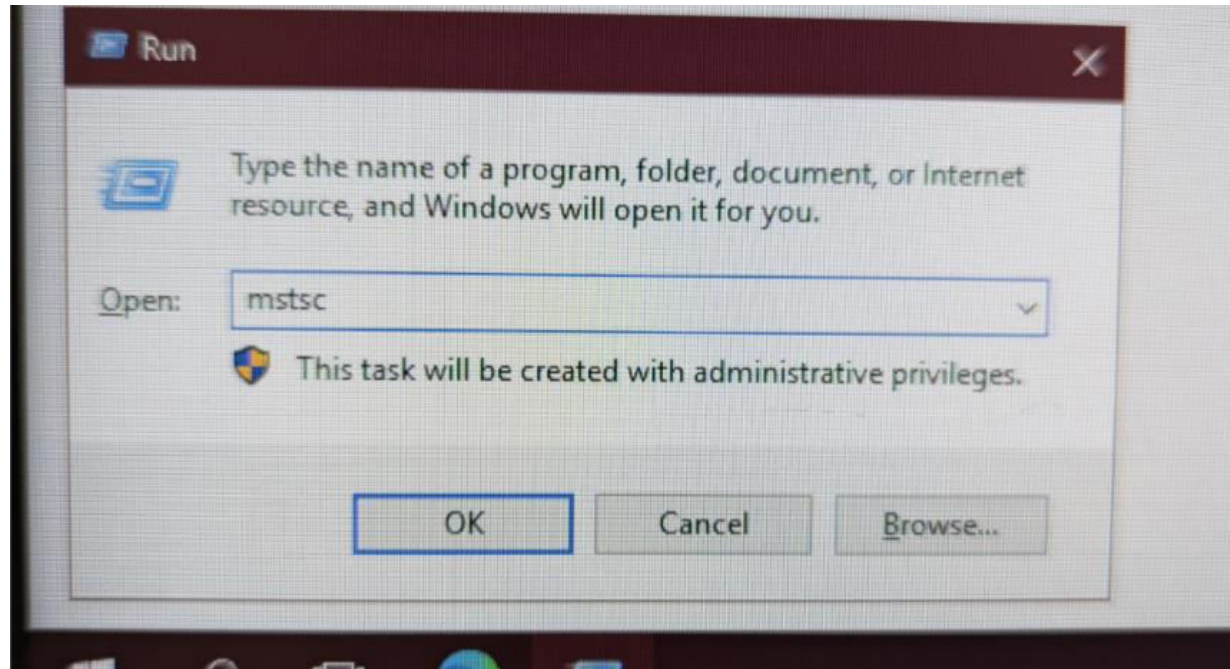
Communication PC - setup



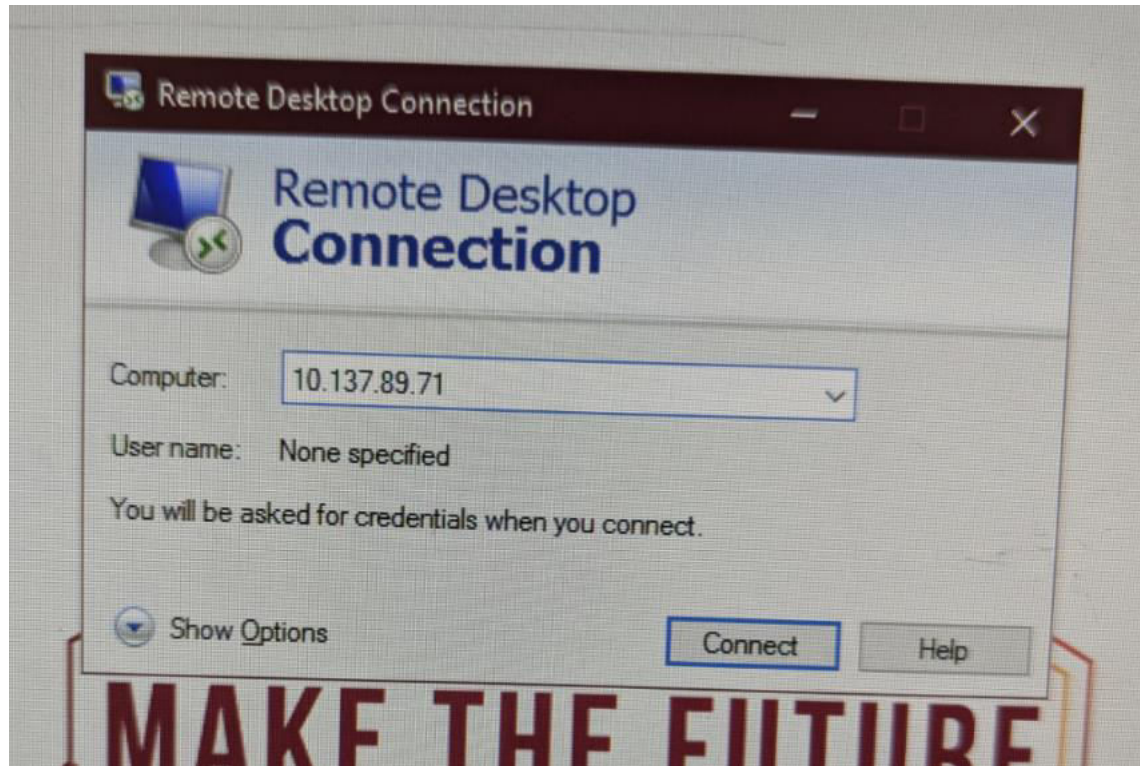
Communication PC – VM_1



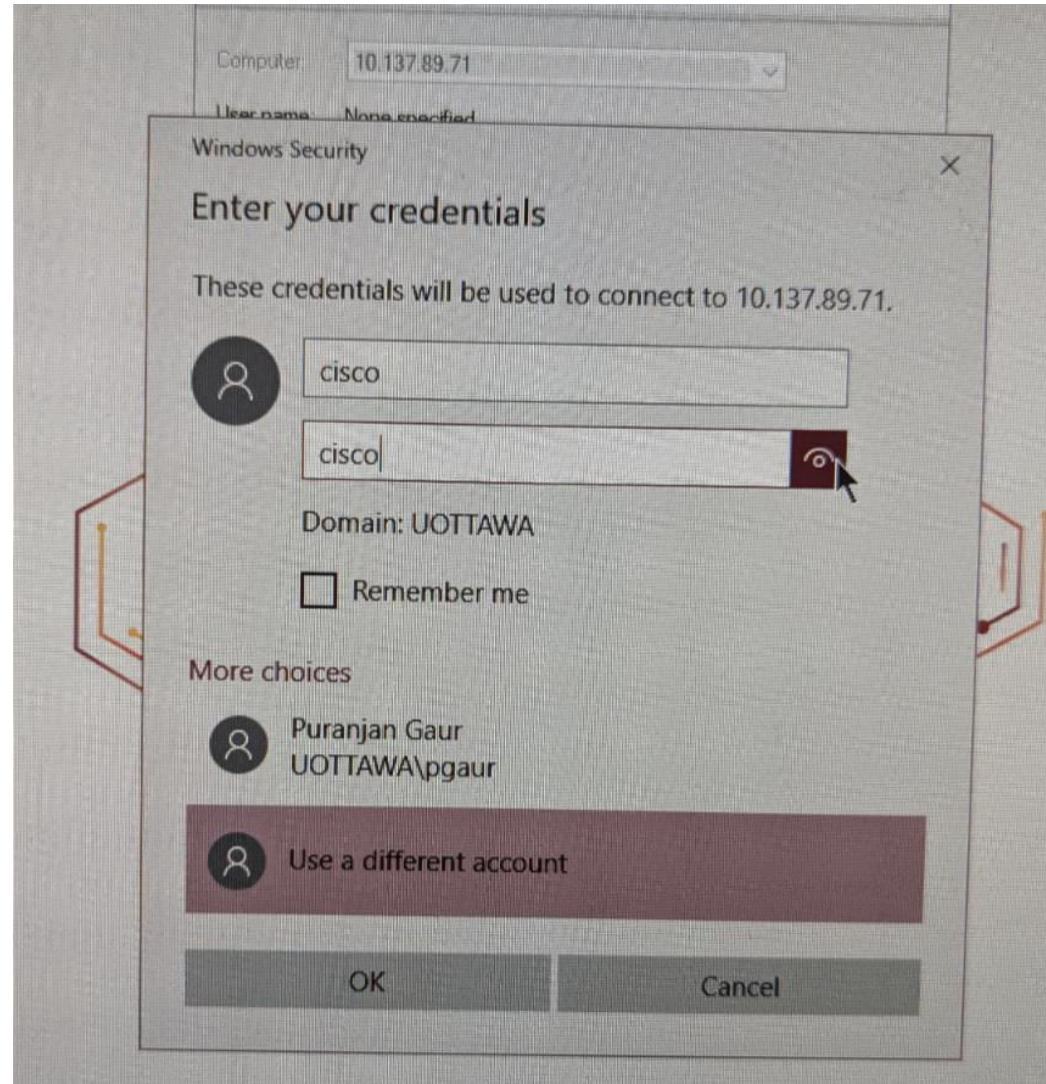
Communication PC – VM_1



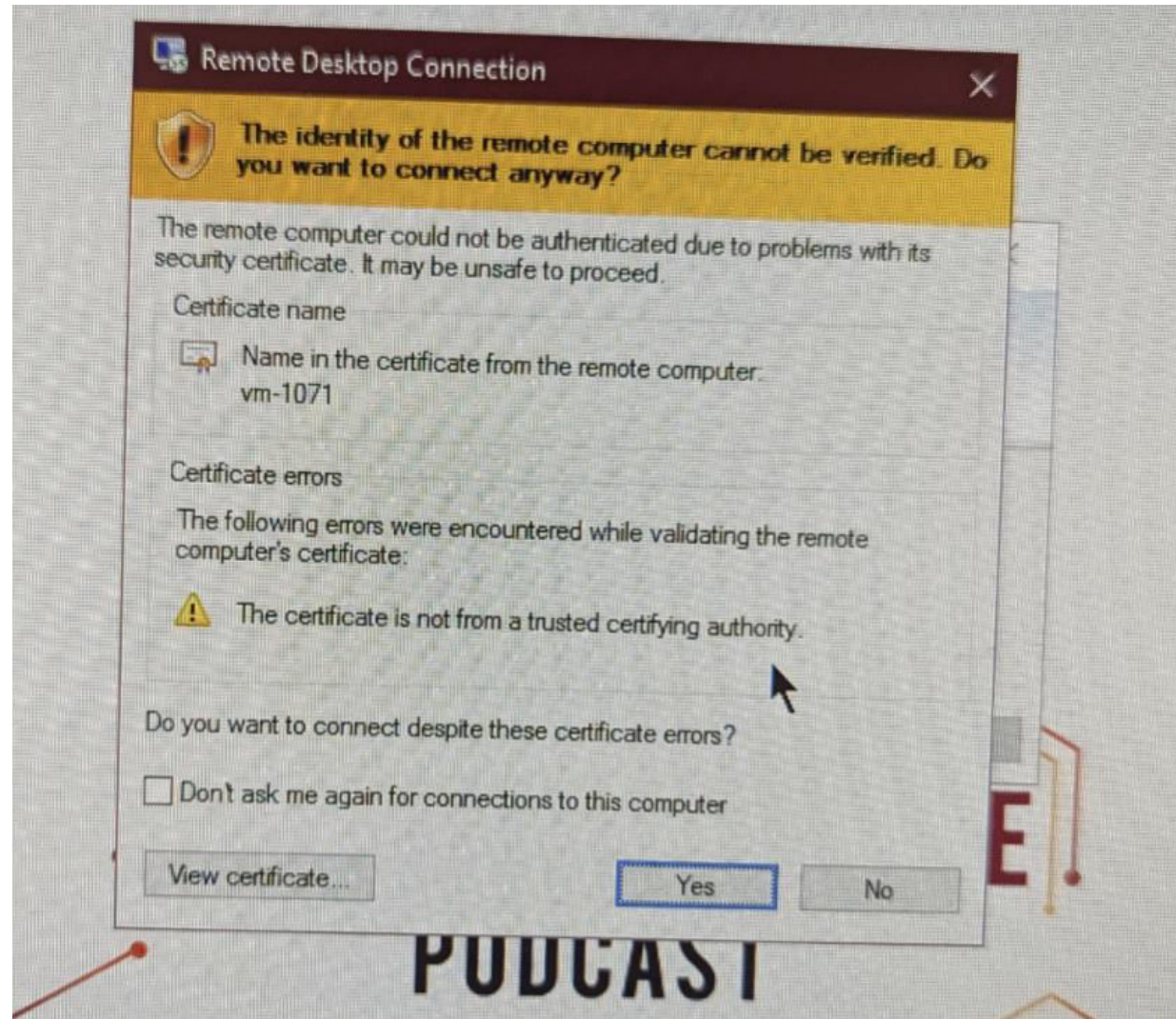
Communication PC – VM_1



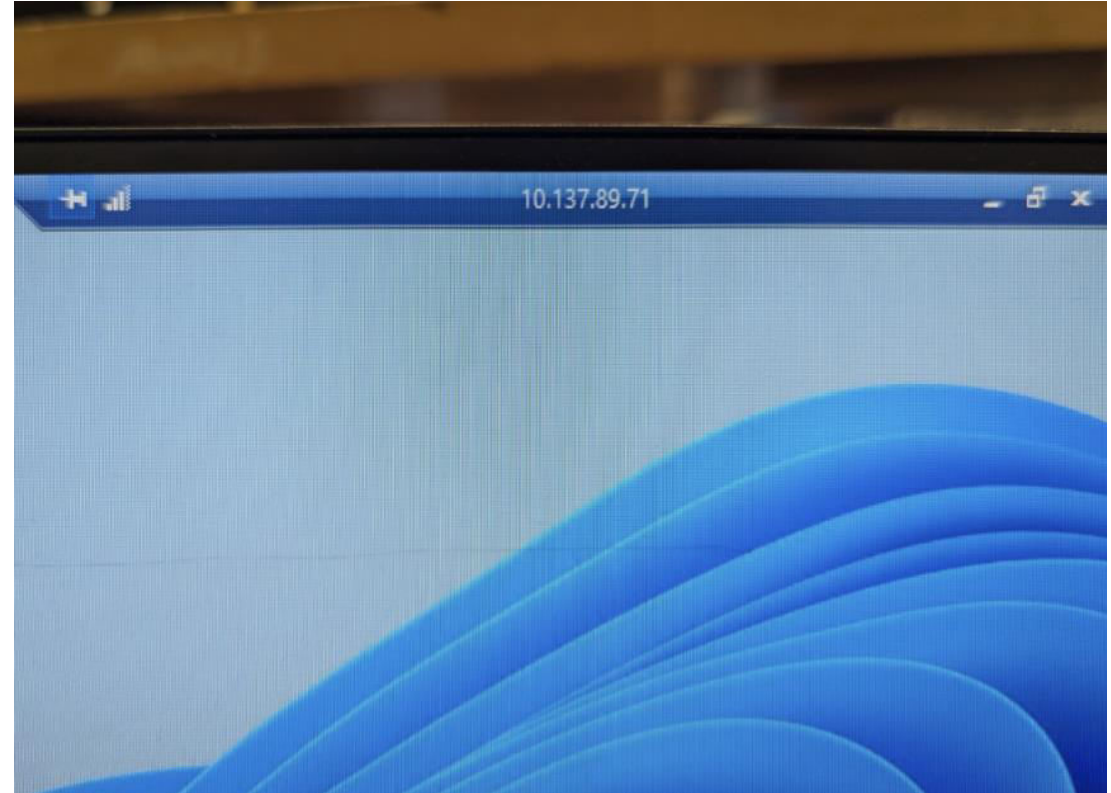
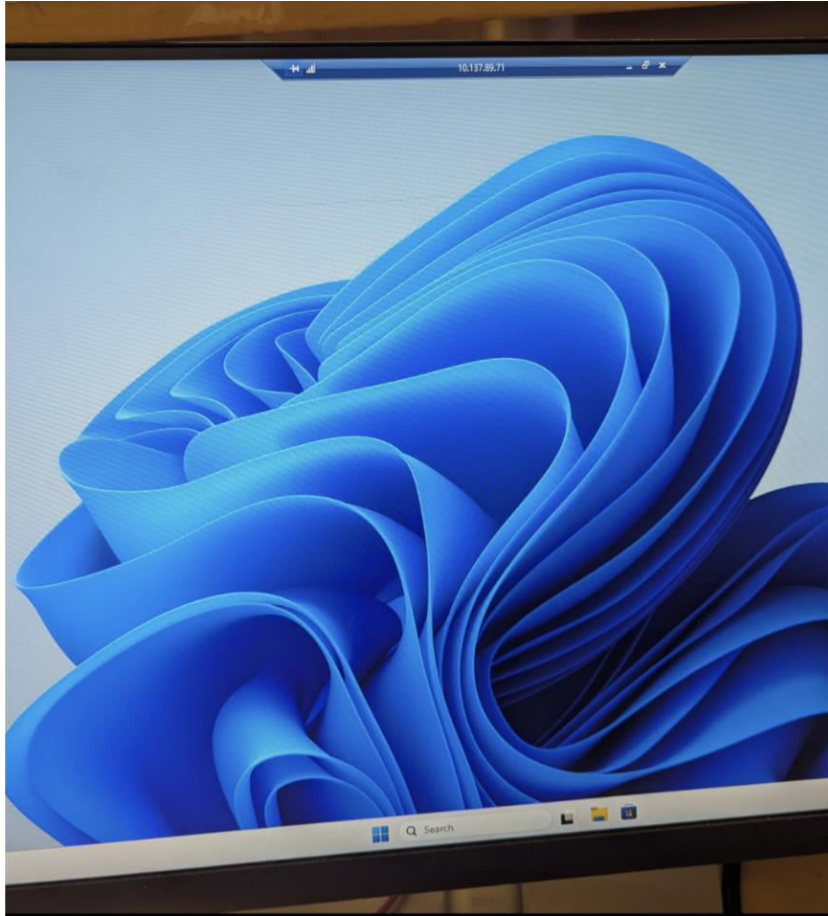
Communication PC – VM_1



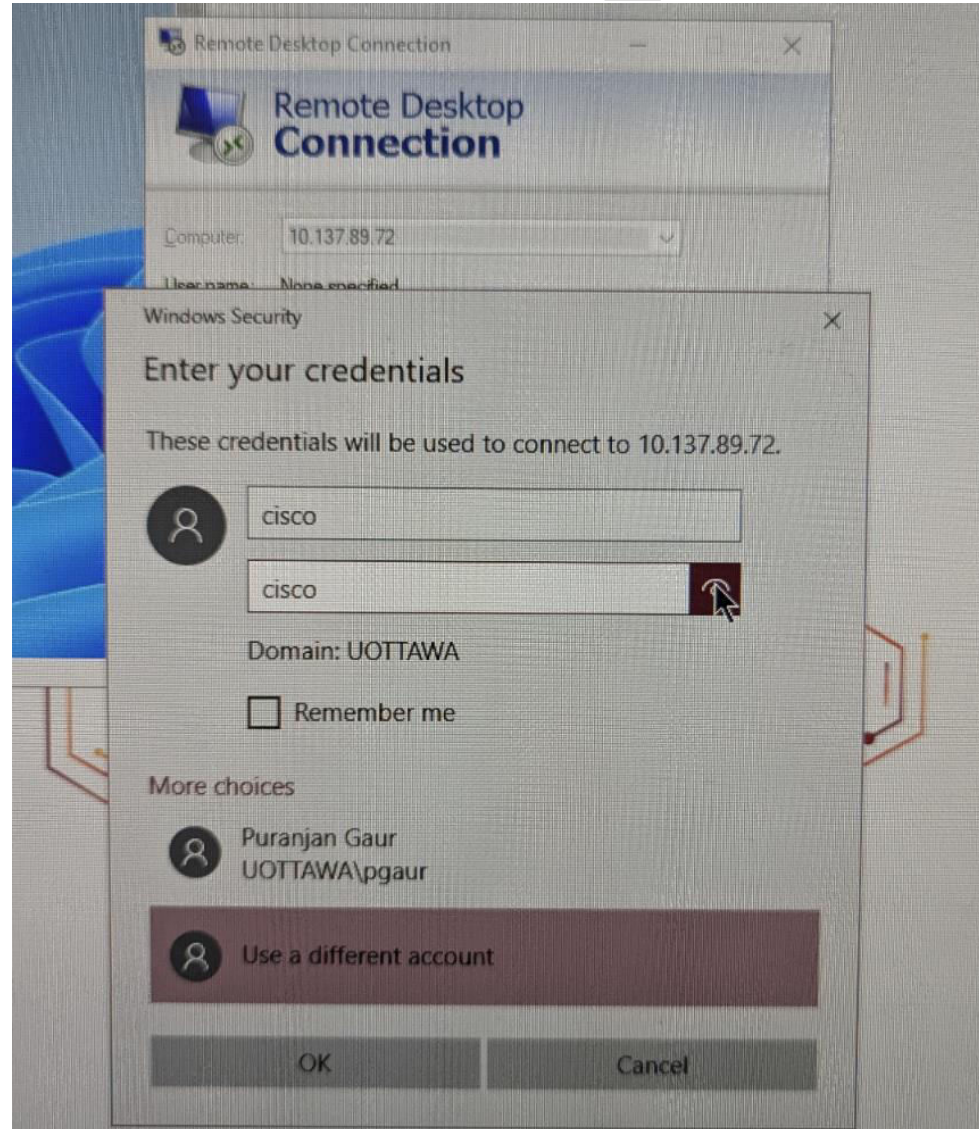
Communication PC – VM_1



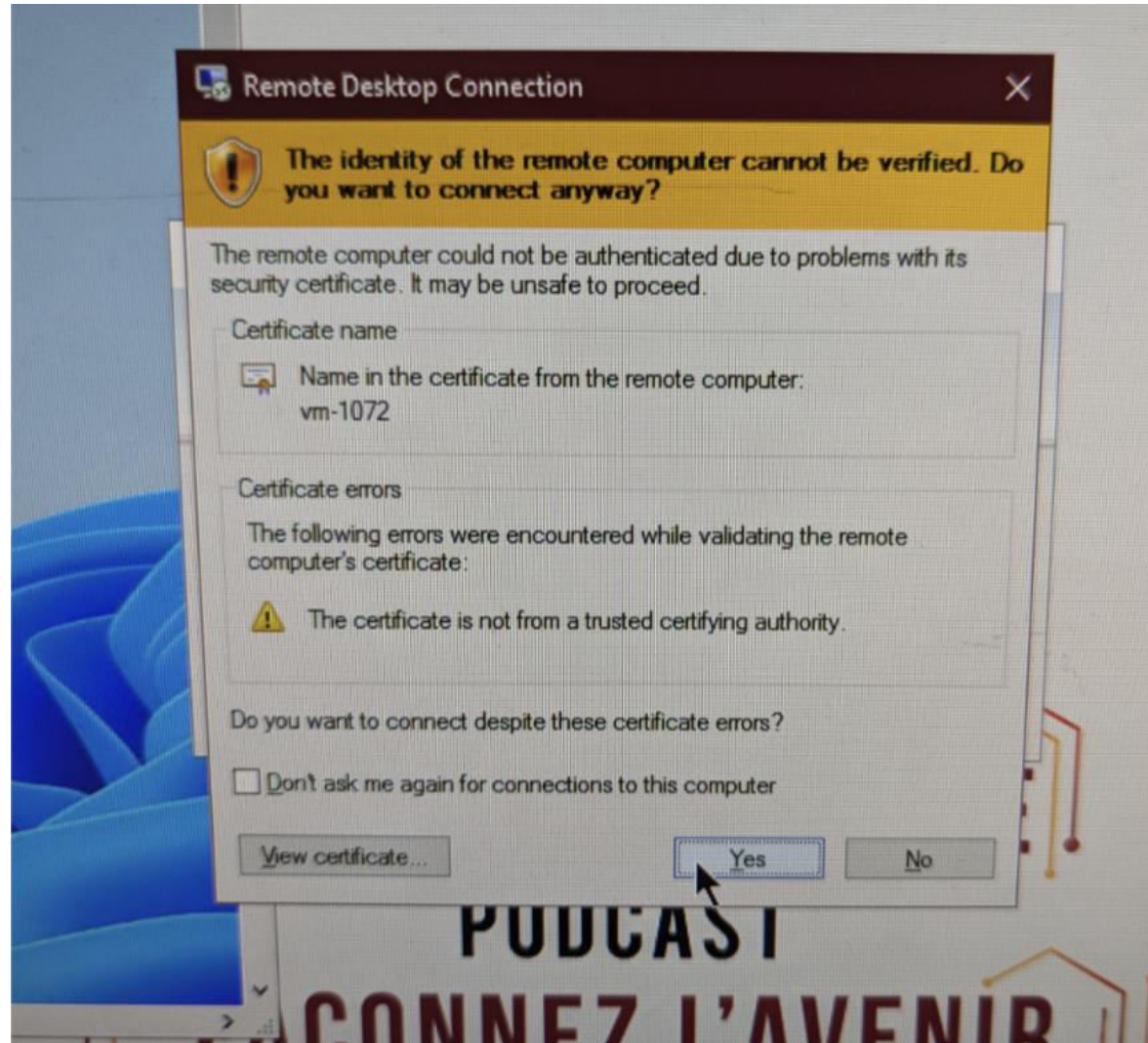
Communication PC – VM_1



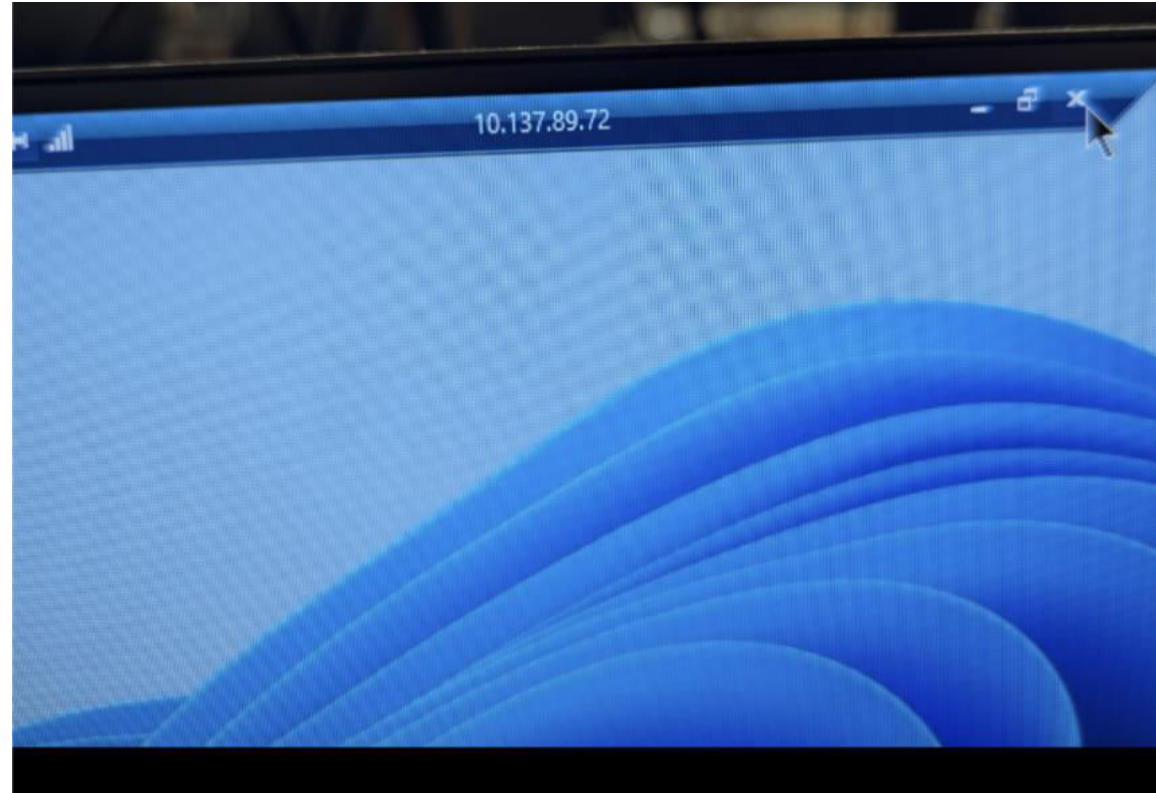
Communication PC – VM_2



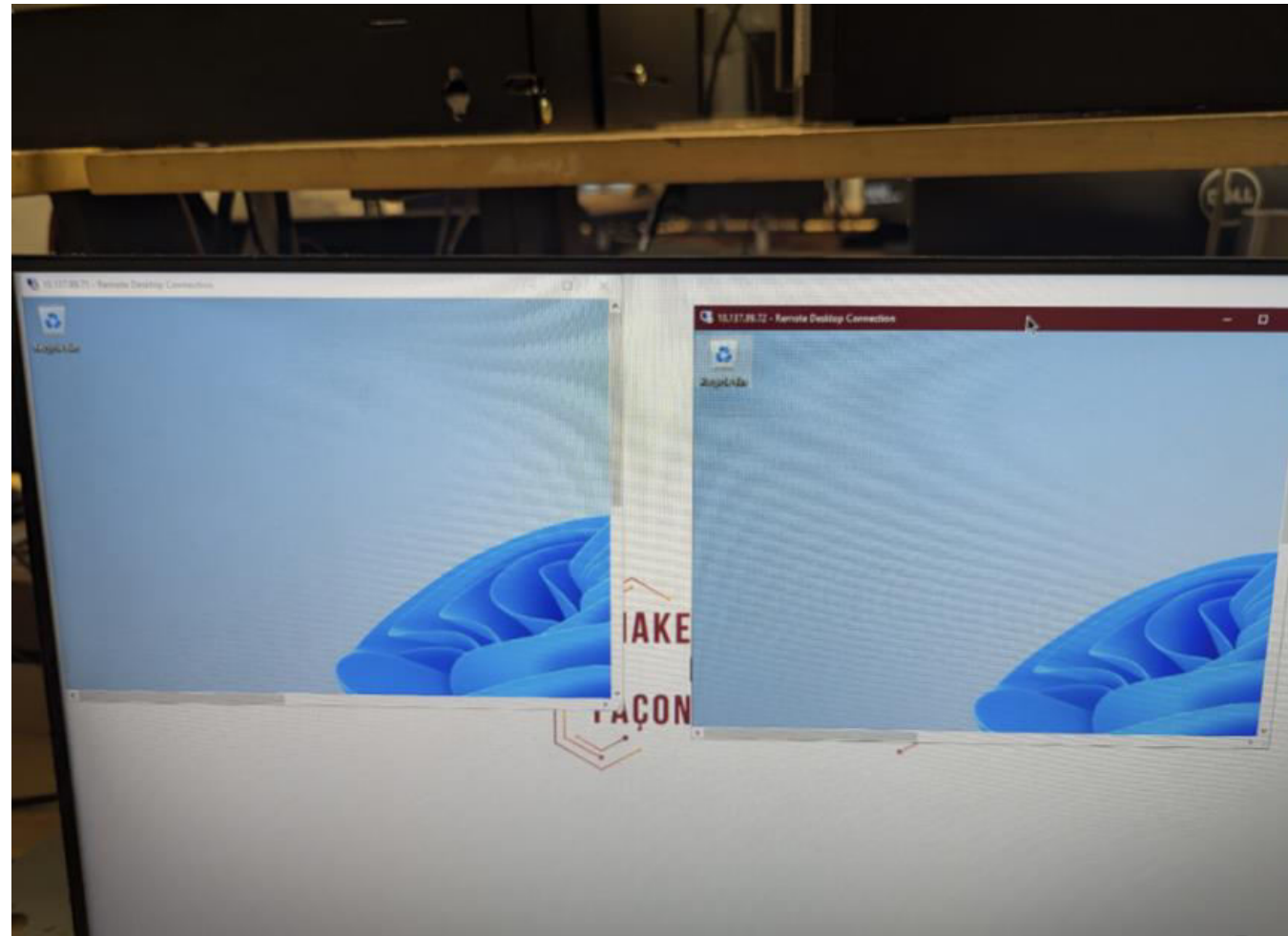
Communication PC – VM_2



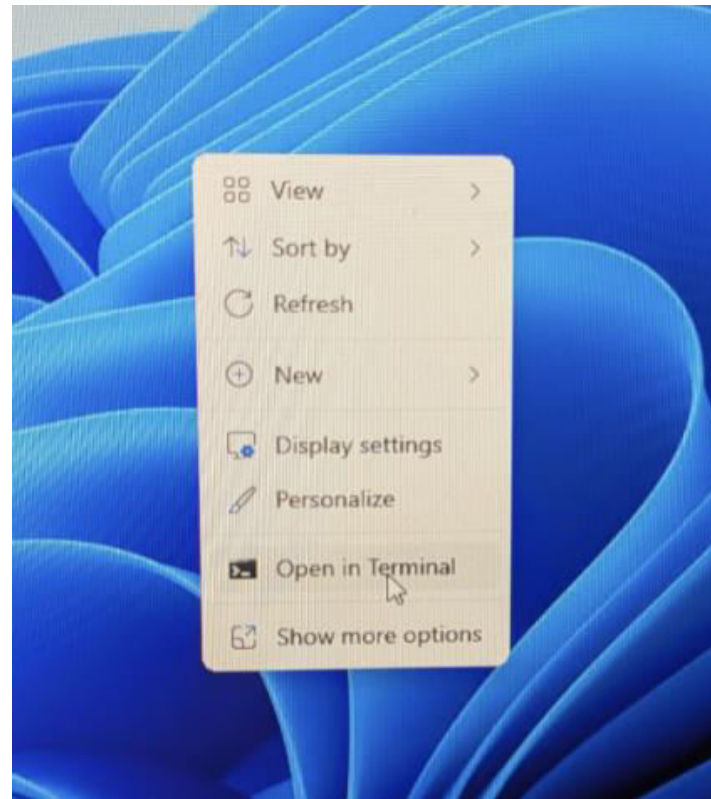
Communication PC – VM_2



Communication PC – VM Display arrangement



Communication PC – VM_1 terminal



Communication PC – VM_1 terminal

```
C:\Users\cisco\Desktop>ipconfig /all

Windows IP Configuration

Host Name . . . . . : vm-1071
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

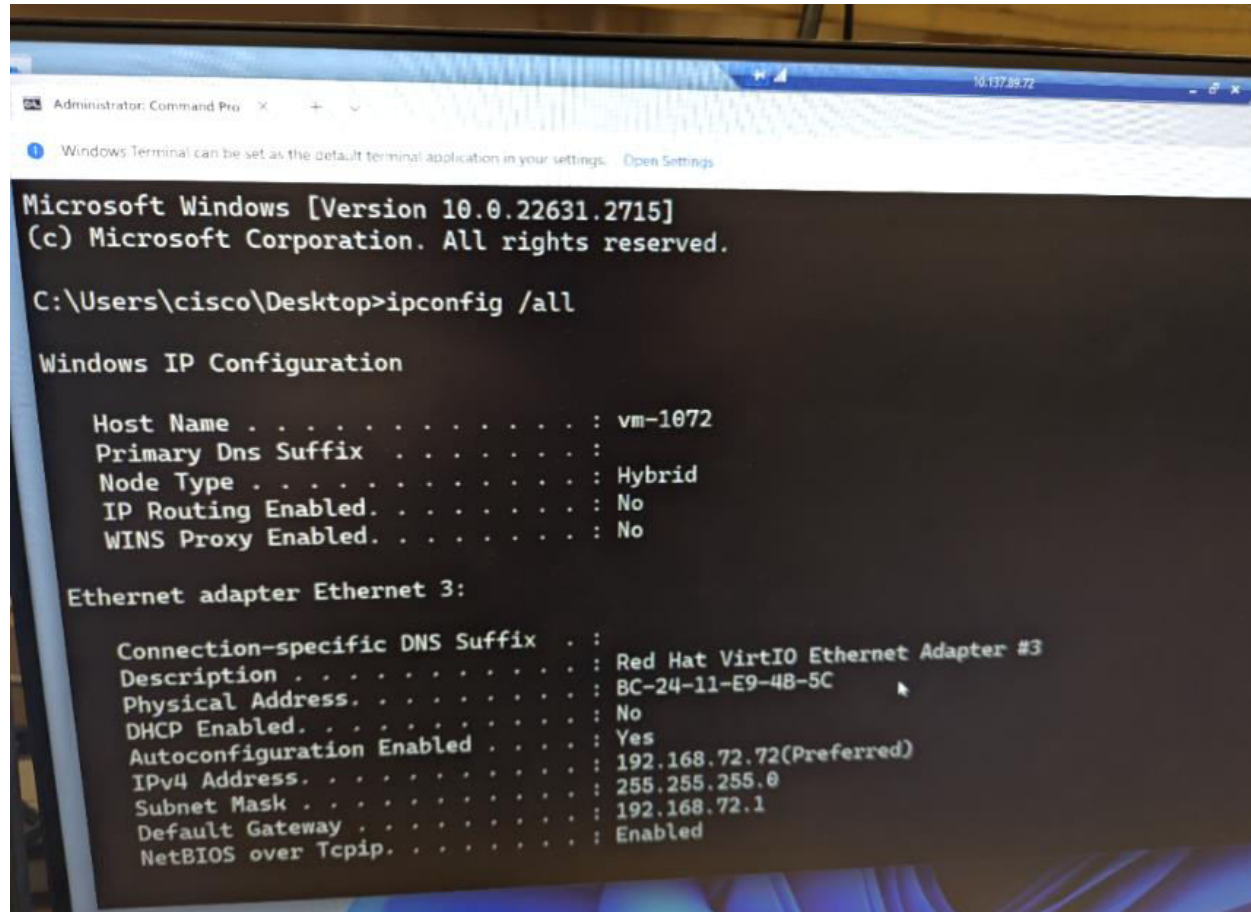
Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : Red Hat VirtIO Ethernet Adapter #3
Physical Address. . . . . : BC-24-11-2B-FB-77
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.71.71(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.71.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : Red Hat VirtIO Ethernet Adapter #2
```

Communication PC – VM_2 terminal



```
Administrator: Command Prom 10.137.88.72
Windows Terminal can be set as the default terminal application in your settings. Open Settings

Microsoft Windows [Version 10.0.22631.2715]
(c) Microsoft Corporation. All rights reserved.

C:\Users\cisco\Desktop>ipconfig /all

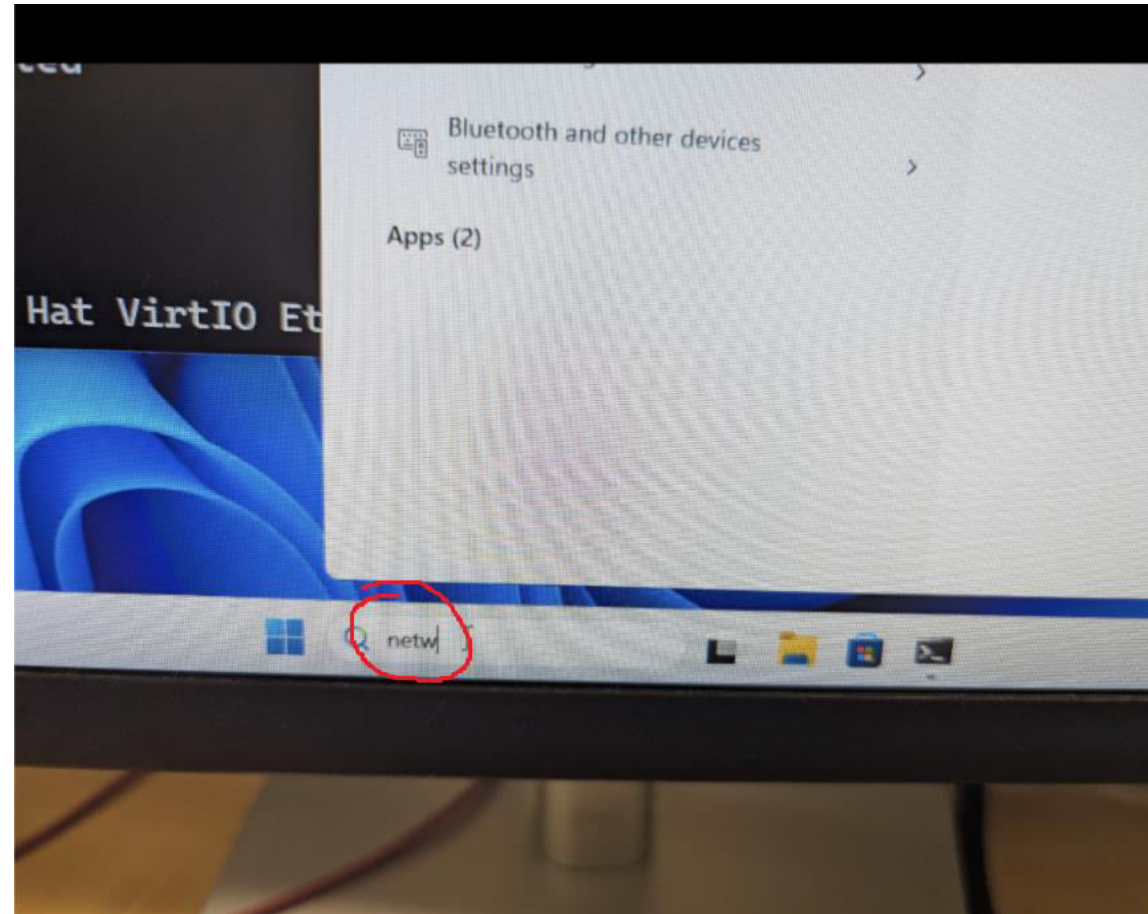
Windows IP Configuration

Host Name . . . . . : vm-1072
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

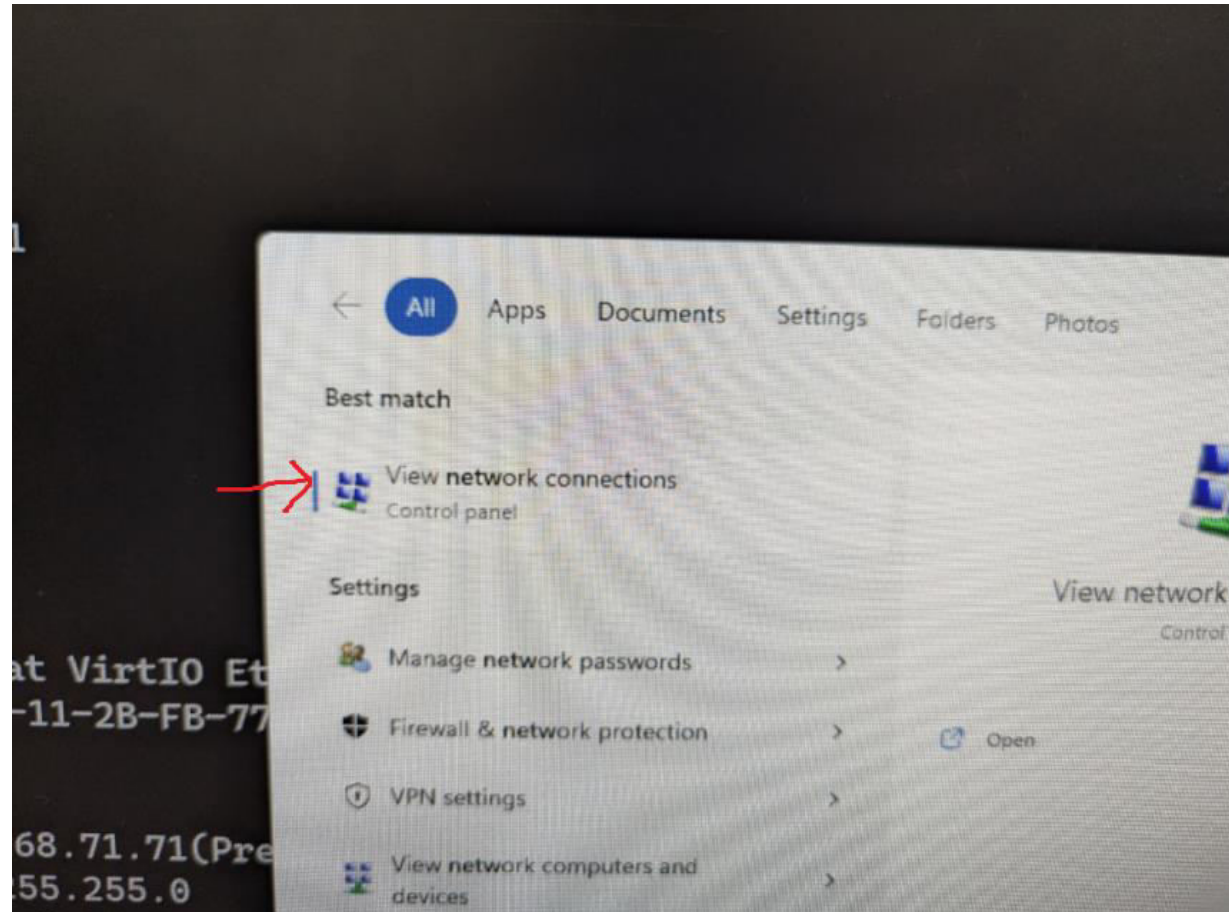
Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : Red Hat VirtIO Ethernet Adapter #3
Physical Address. . . . . : BC-24-11-E9-4B-5C
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . : Yes
IPv4 Address. . . . . : 192.168.72.72(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.72.1
NetBIOS over Tcpip. . . . . : Enabled
```

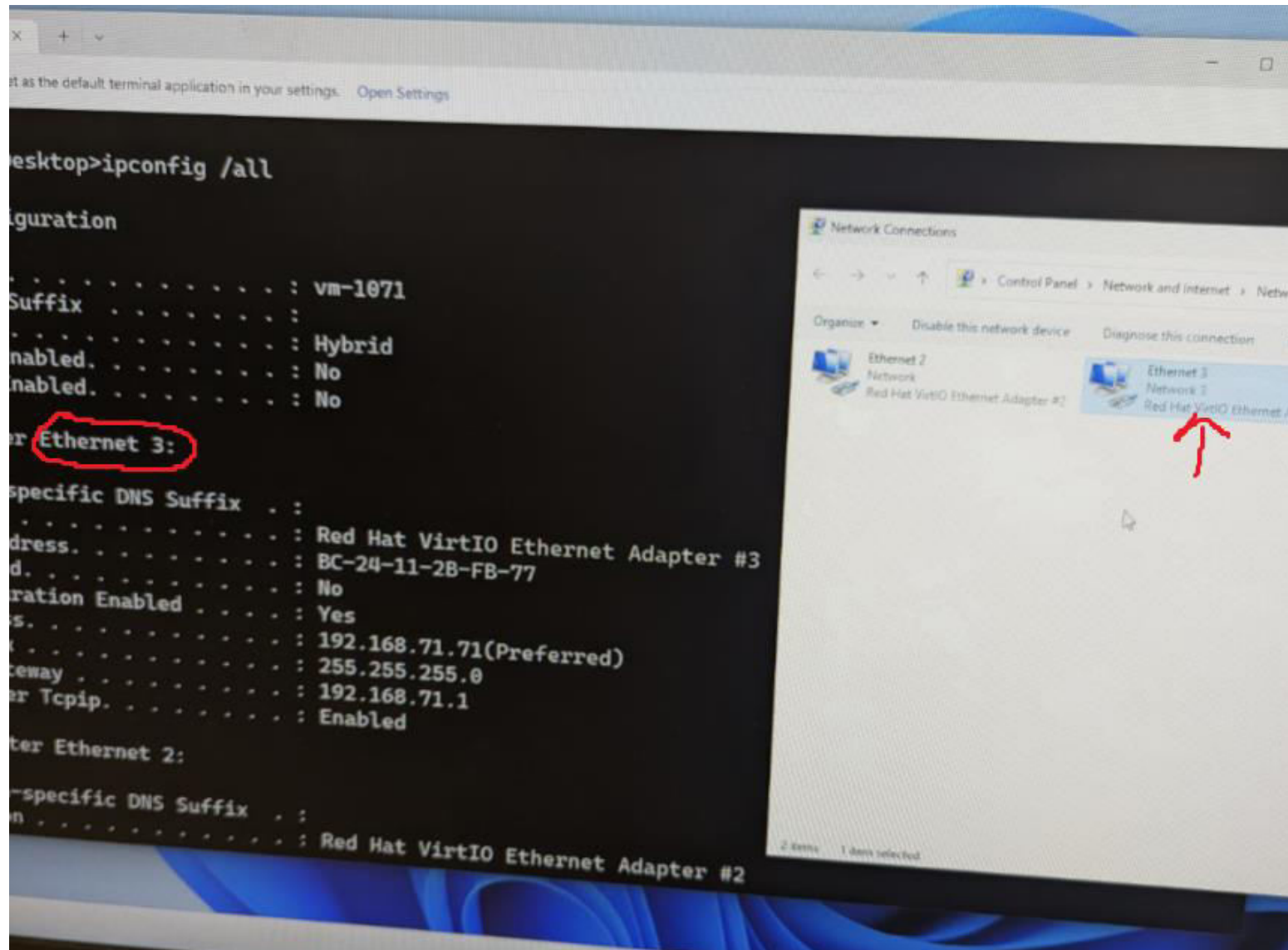
Communication PC – VM_1 Network Interface setting



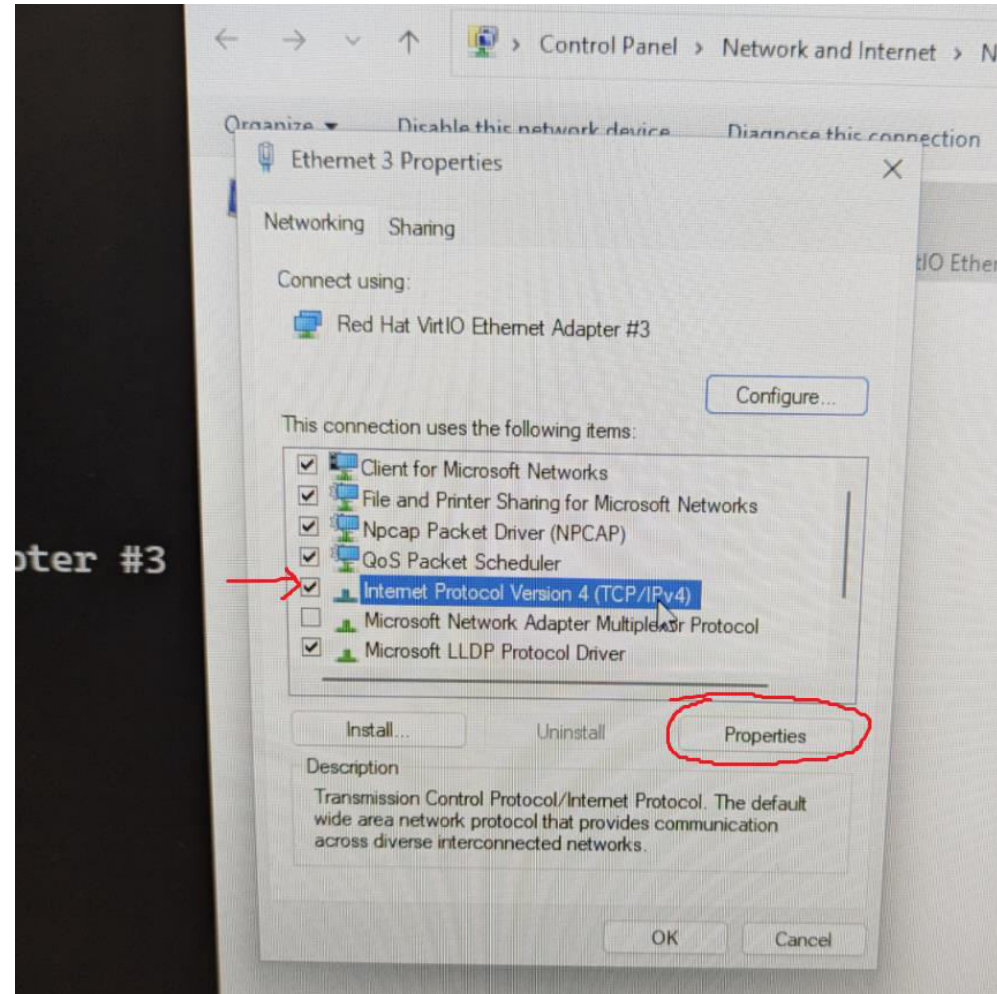
Communication PC – VM_1 NIC configure



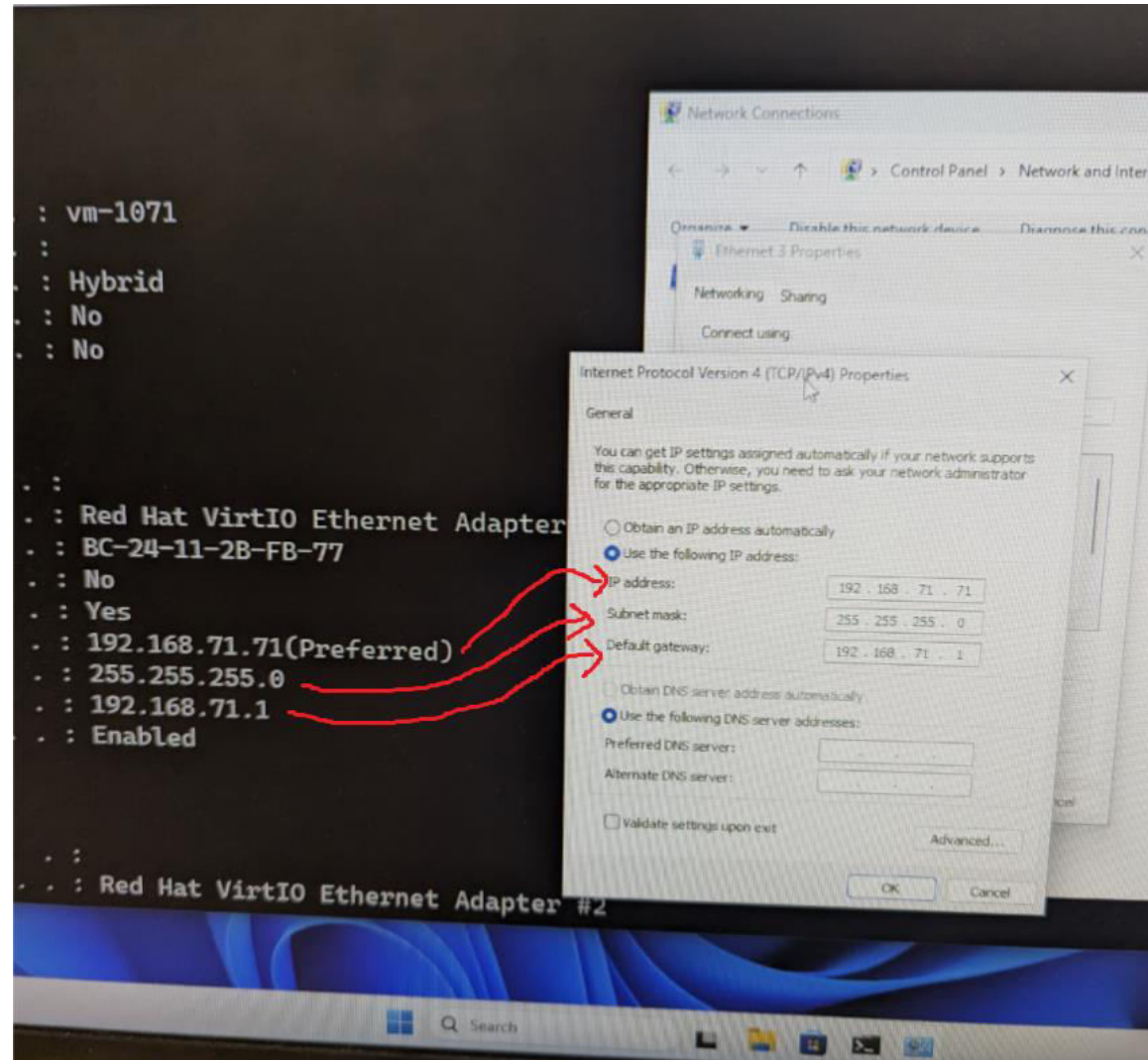
Communication PC – VM 1 NIC configure



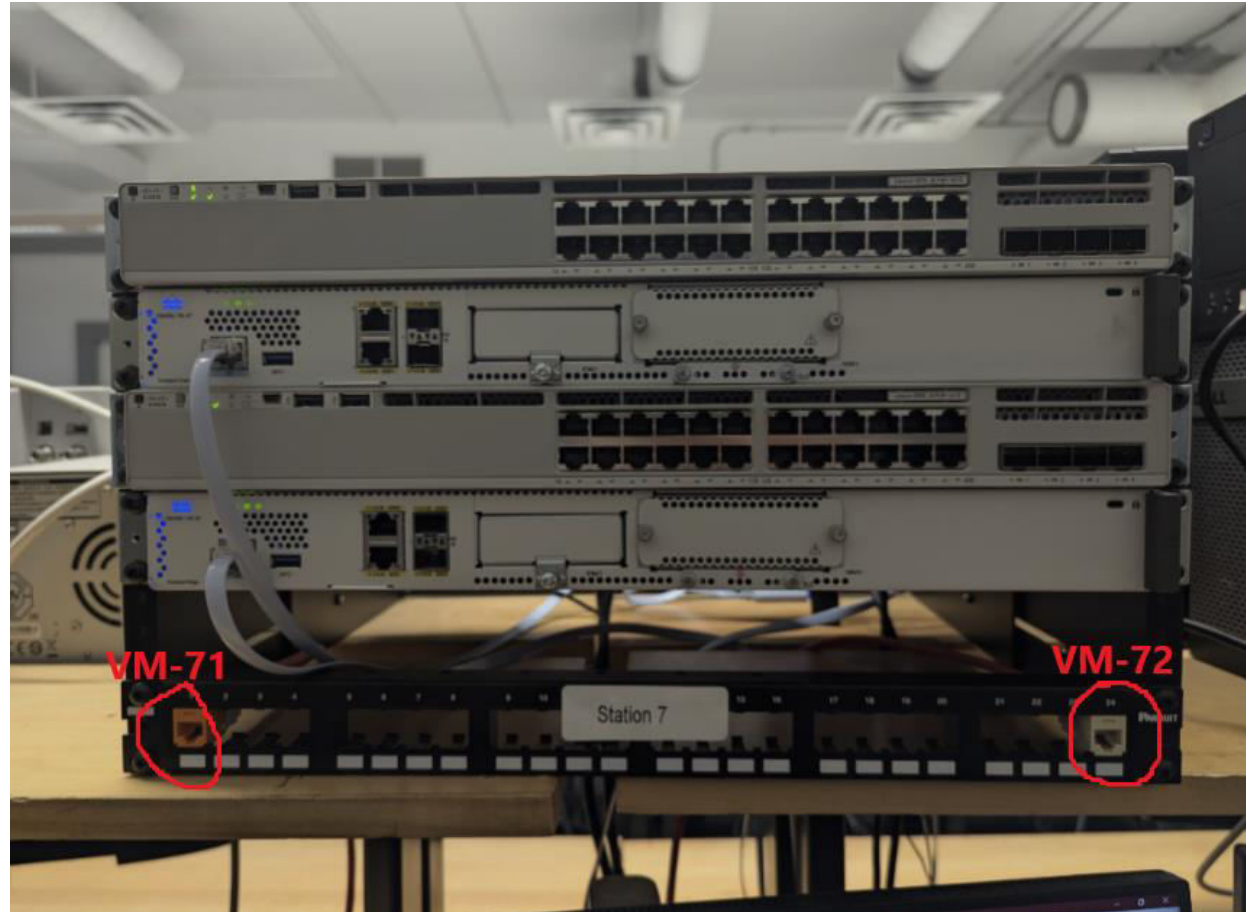
Communication PC – VM_1 NIC configure



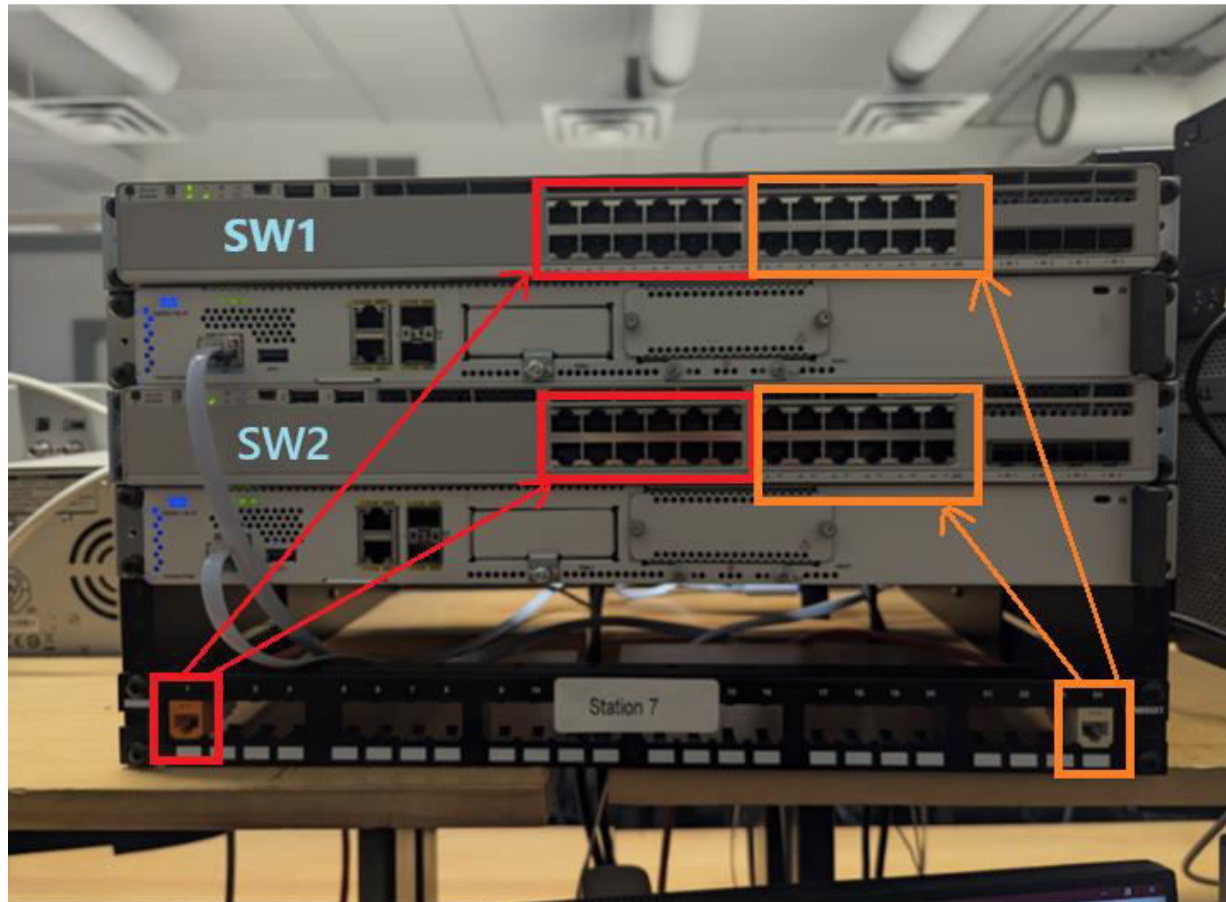
Communication PC – VM_1 NIC Configure



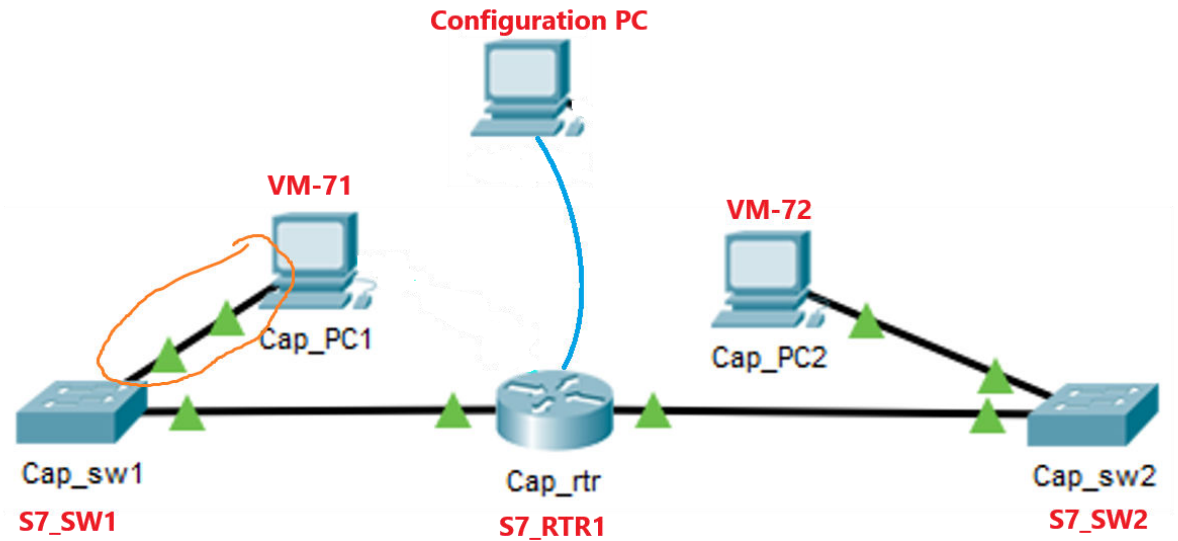
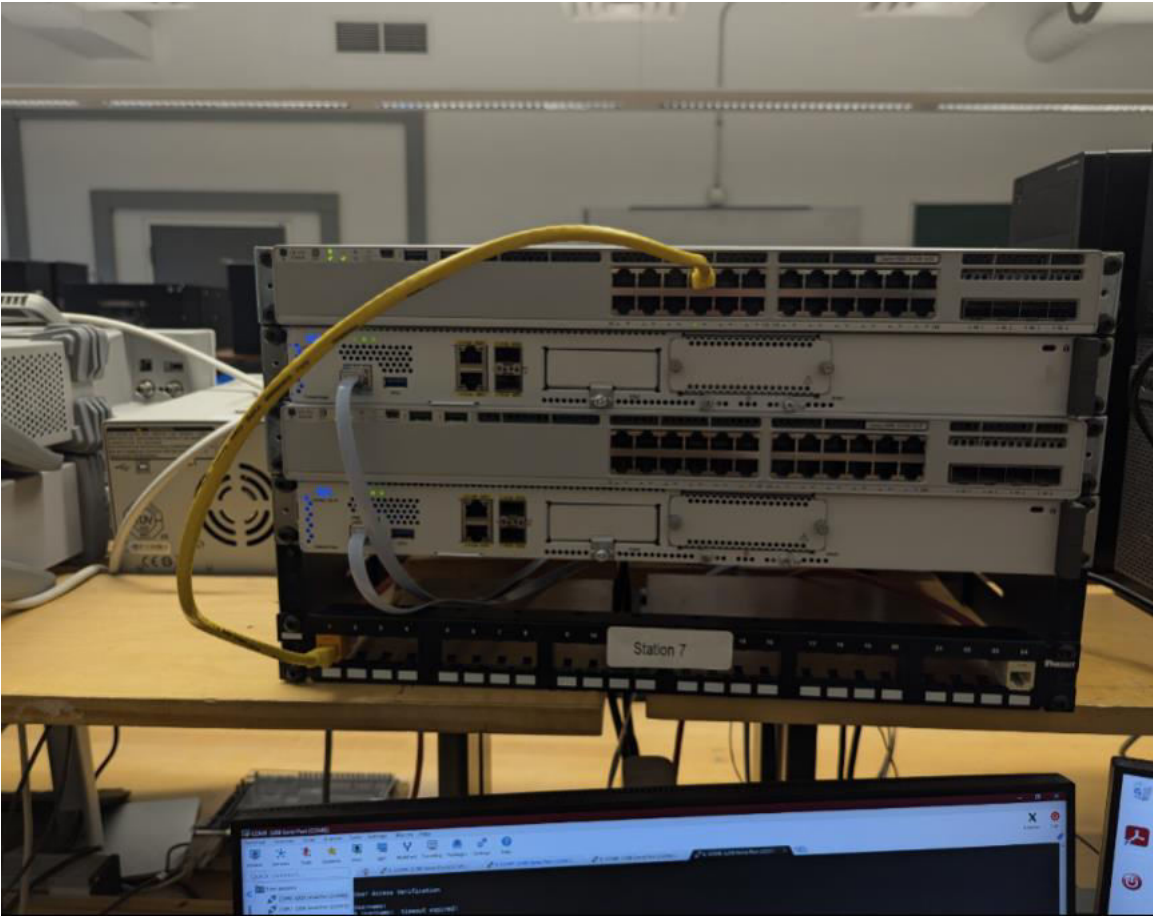
Connecting VMs and Network devices



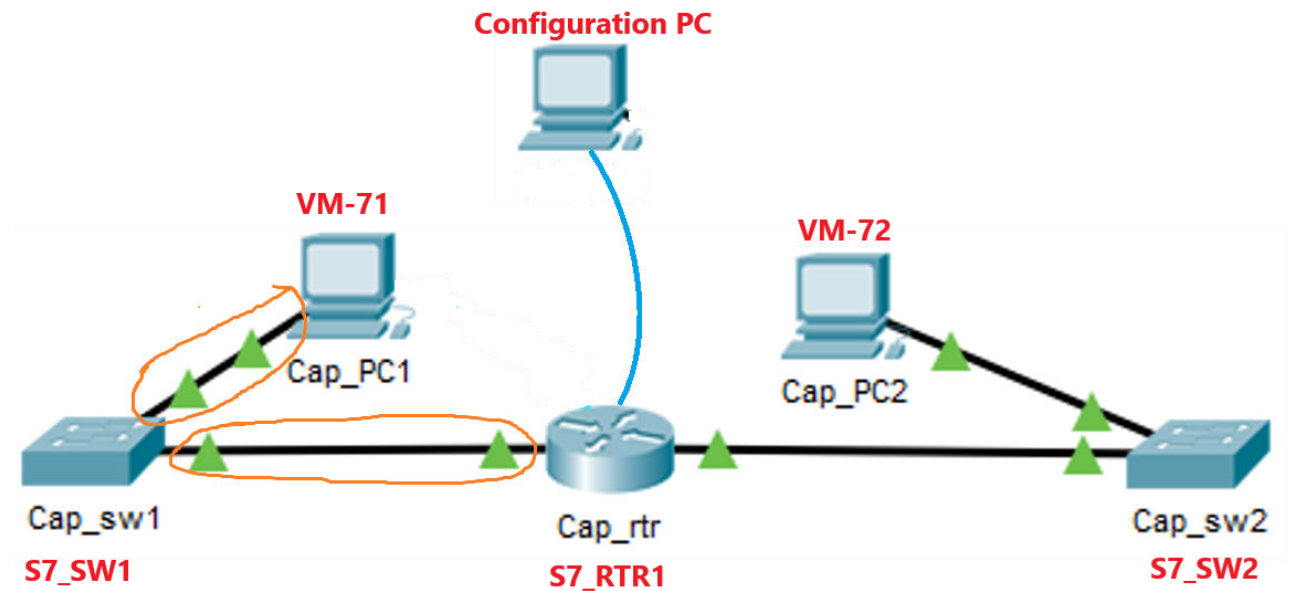
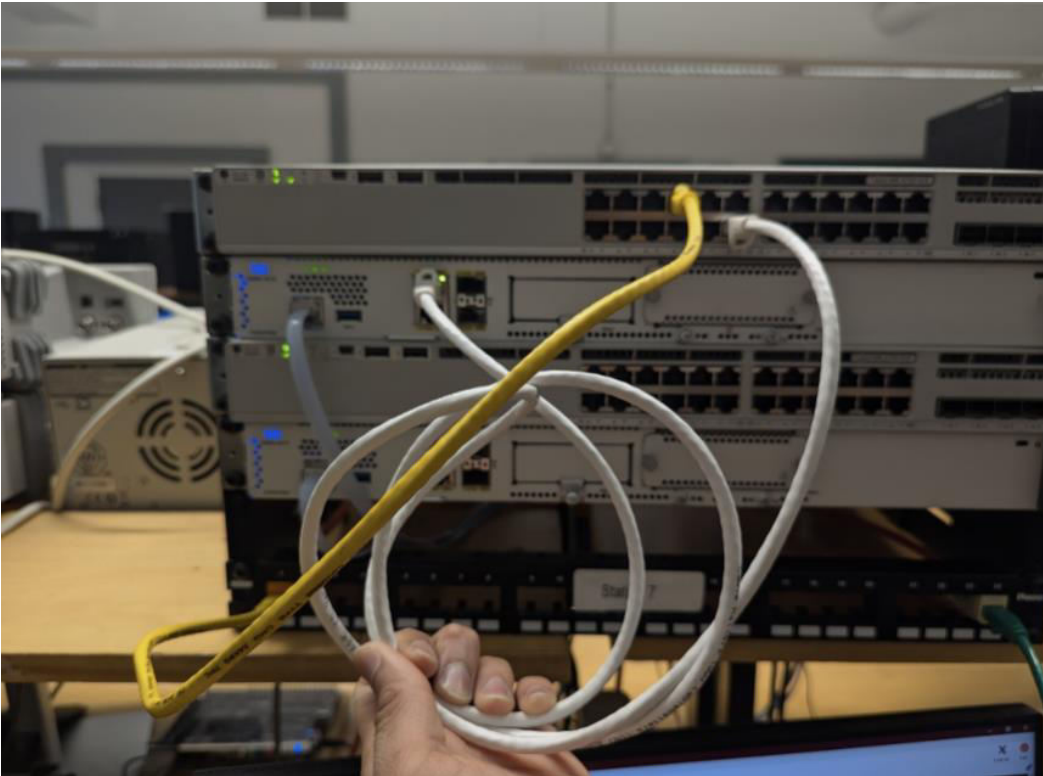
Switch interface connection with VM



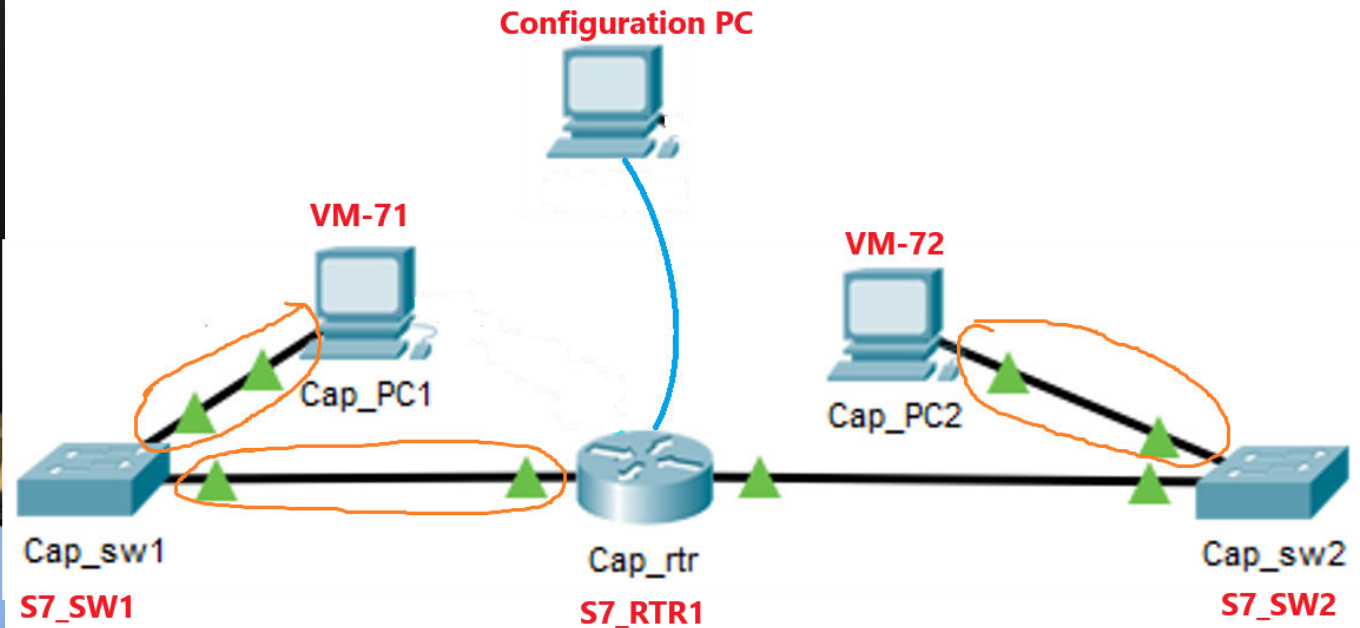
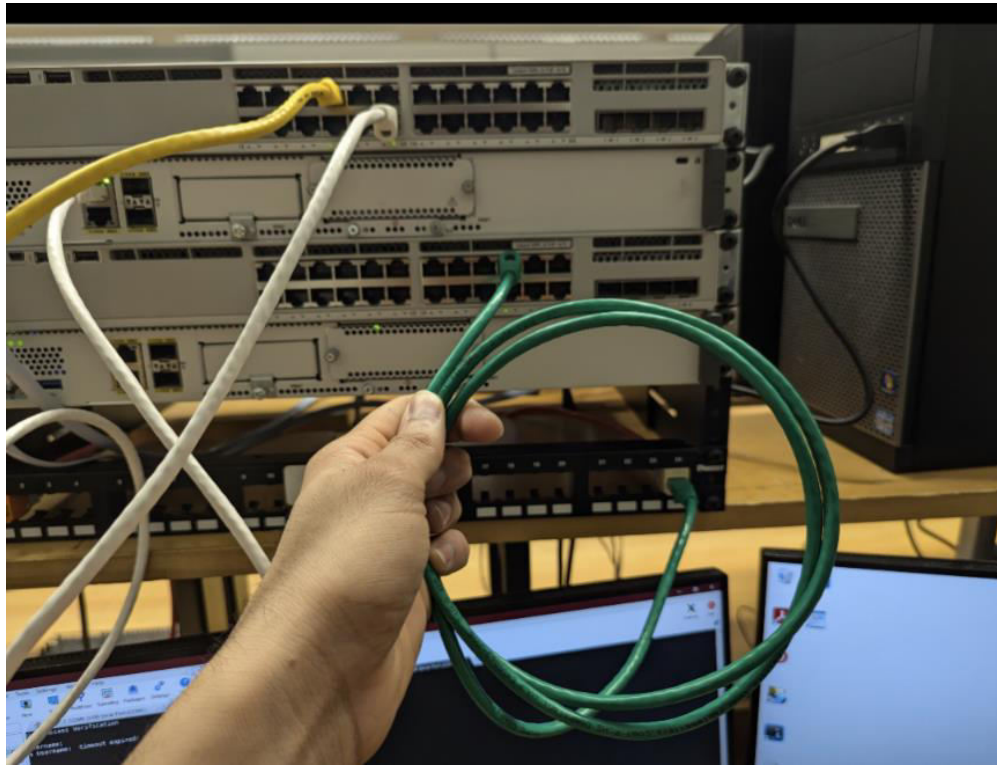
Connecting VMs and Network devices



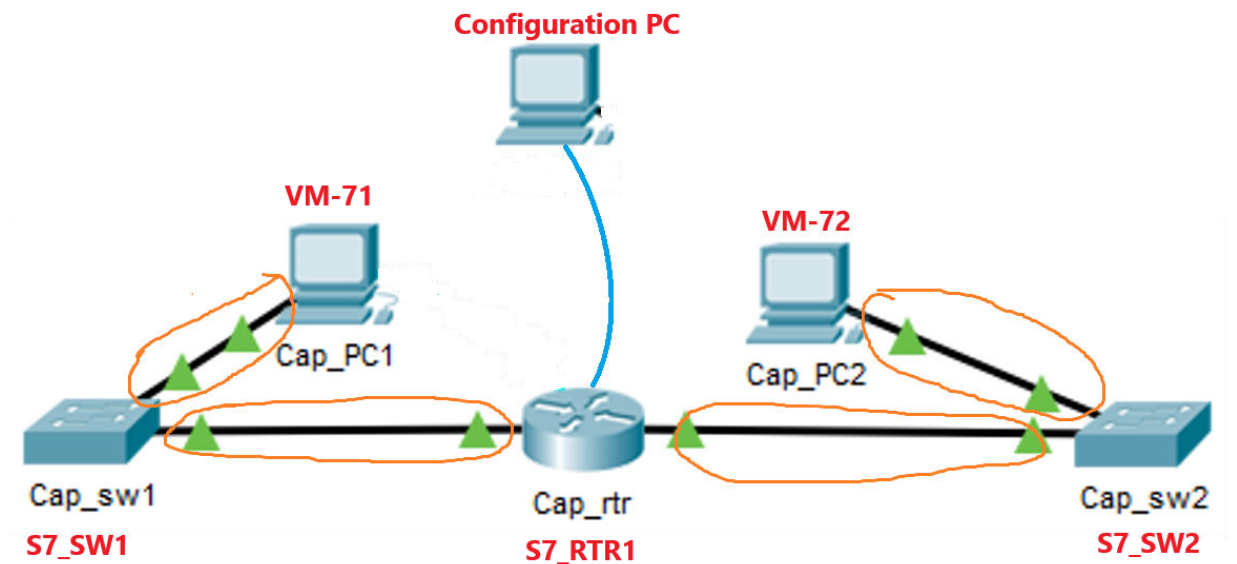
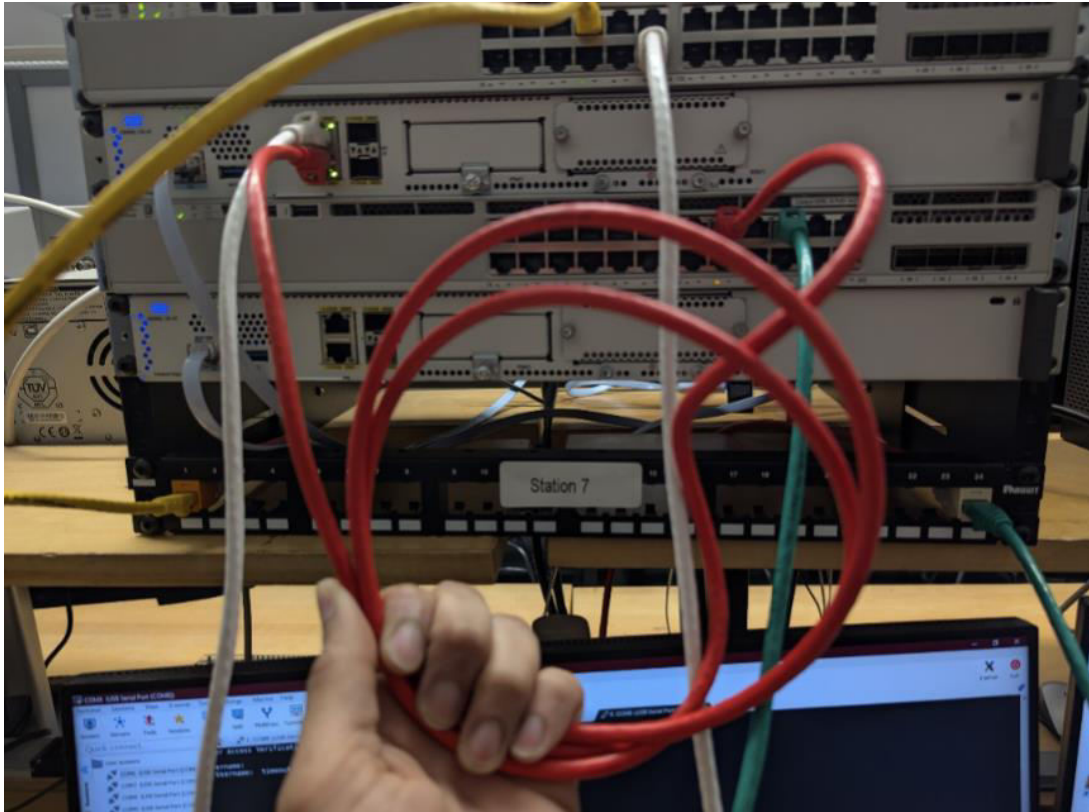
Connecting VMs and Network devices



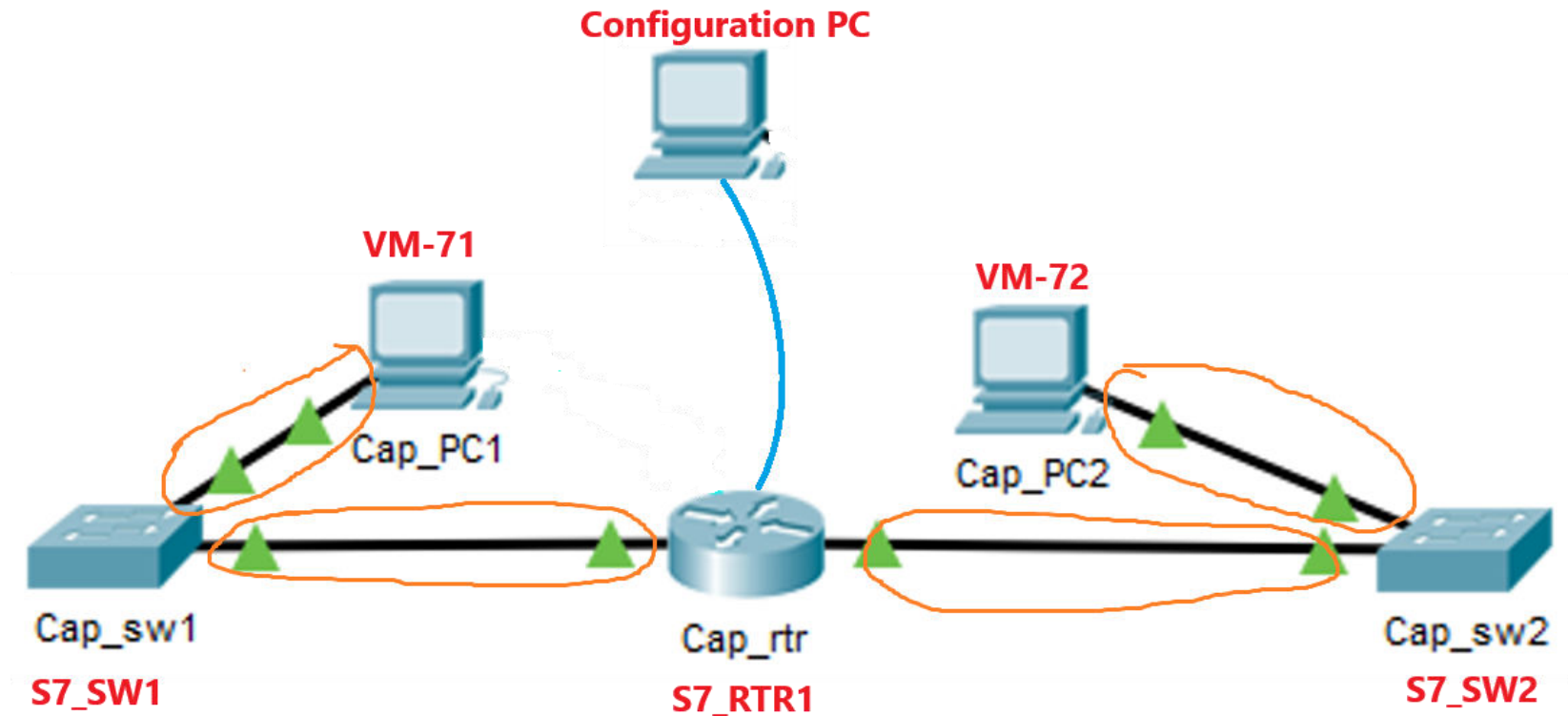
Connecting VMs and Network devices



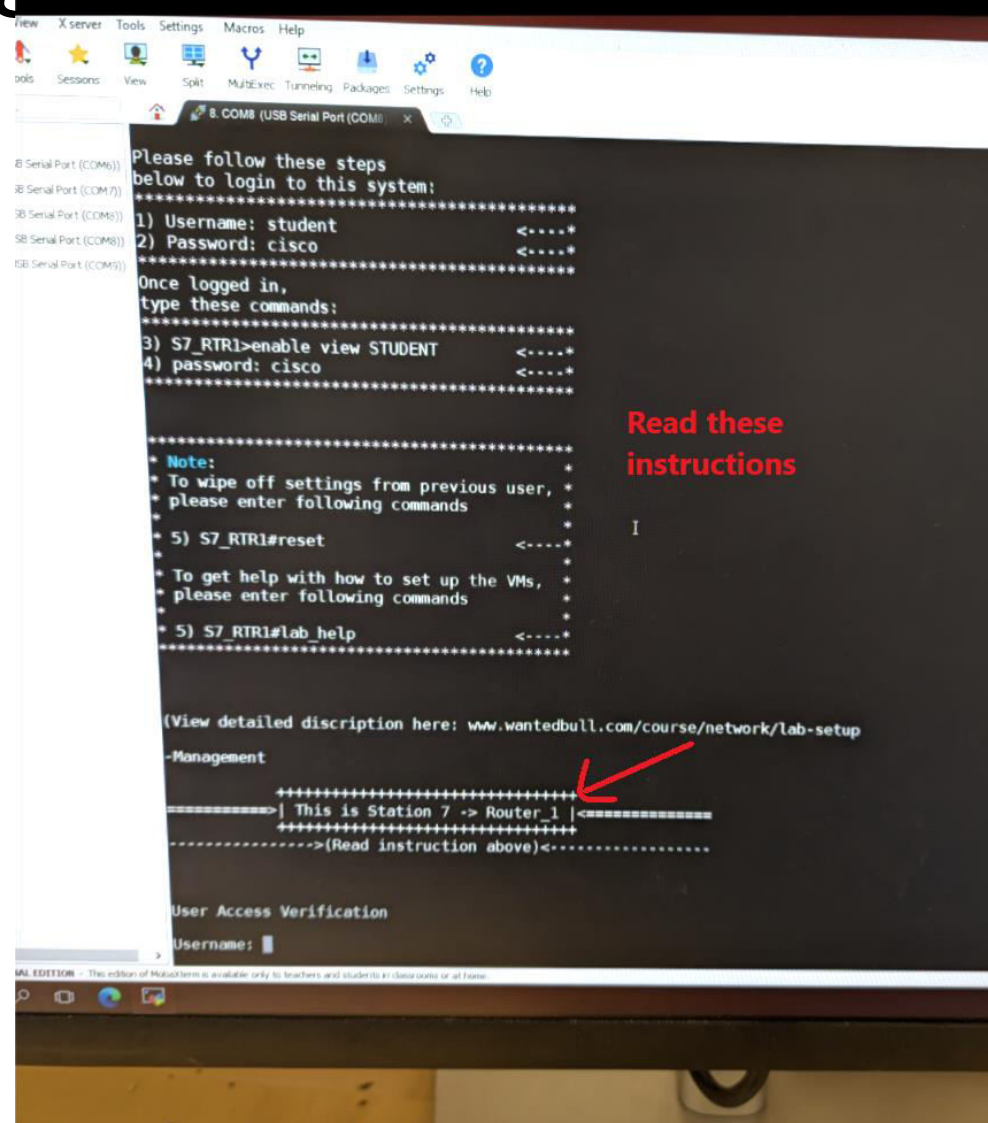
Connecting VMs and Network devices



Connecting VMs and Network devices



Connecting VMs and Network devices



The screenshot shows a terminal window titled "8. COM8 (USB Serial Port (COM8))". The terminal displays a series of instructions for logging into a system. The instructions are as follows:

```
Please follow these steps
below to login to this system:
*****
1) Username: student          <----*
2) Password: cisco           <----*
*****
Once logged in,
type these commands:
*****
3) S7_RTR1>enable view STUDENT <----*
4) password: cisco            <----*
*****

* Note:
* To wipe off settings from previous user,
* please enter following commands
*
* 5) S7_RTR1#reset             <----*
*
* To get help with how to set up the VMs,
* please enter following commands
*
* 5) S7_RTR1#lab_help          <----*
*****

(View detailed discription here: www.wantedbull.com/course/network/lab-setup)

-Management

+++++
=====| This is Station 7 -> Router_1 |=====
+++++
----->(Read instruction above)<-----

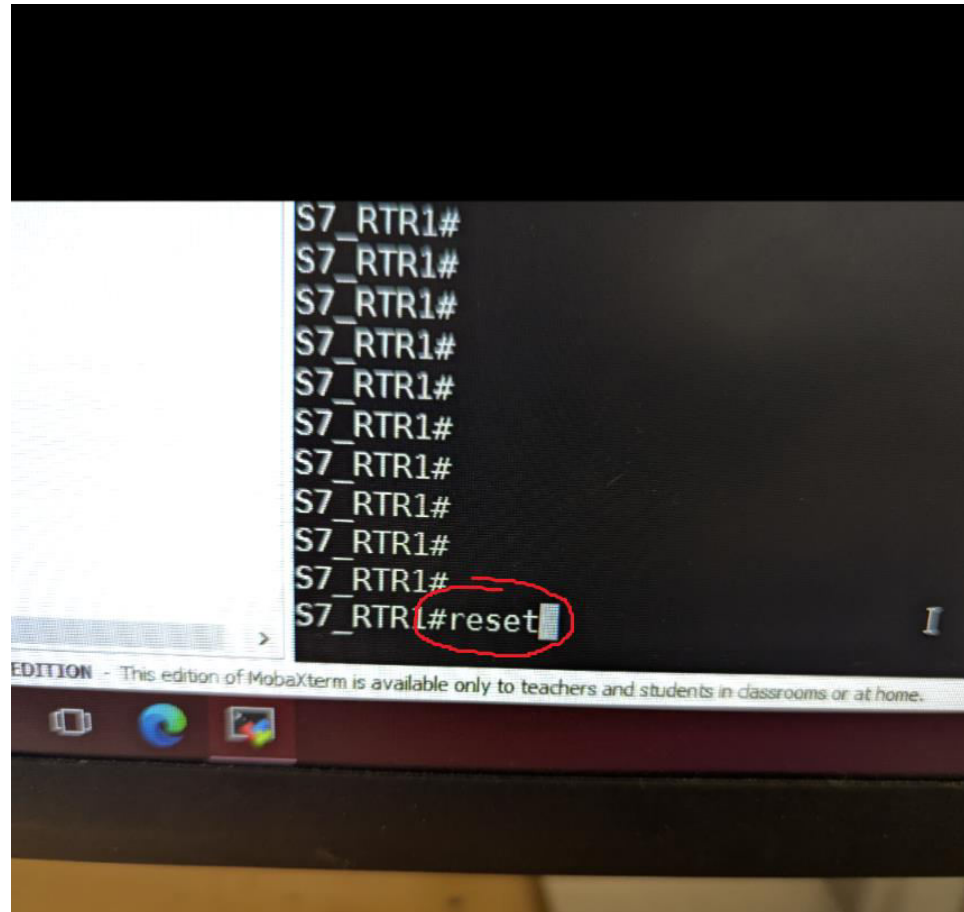
User Access Verification

Username: 
```

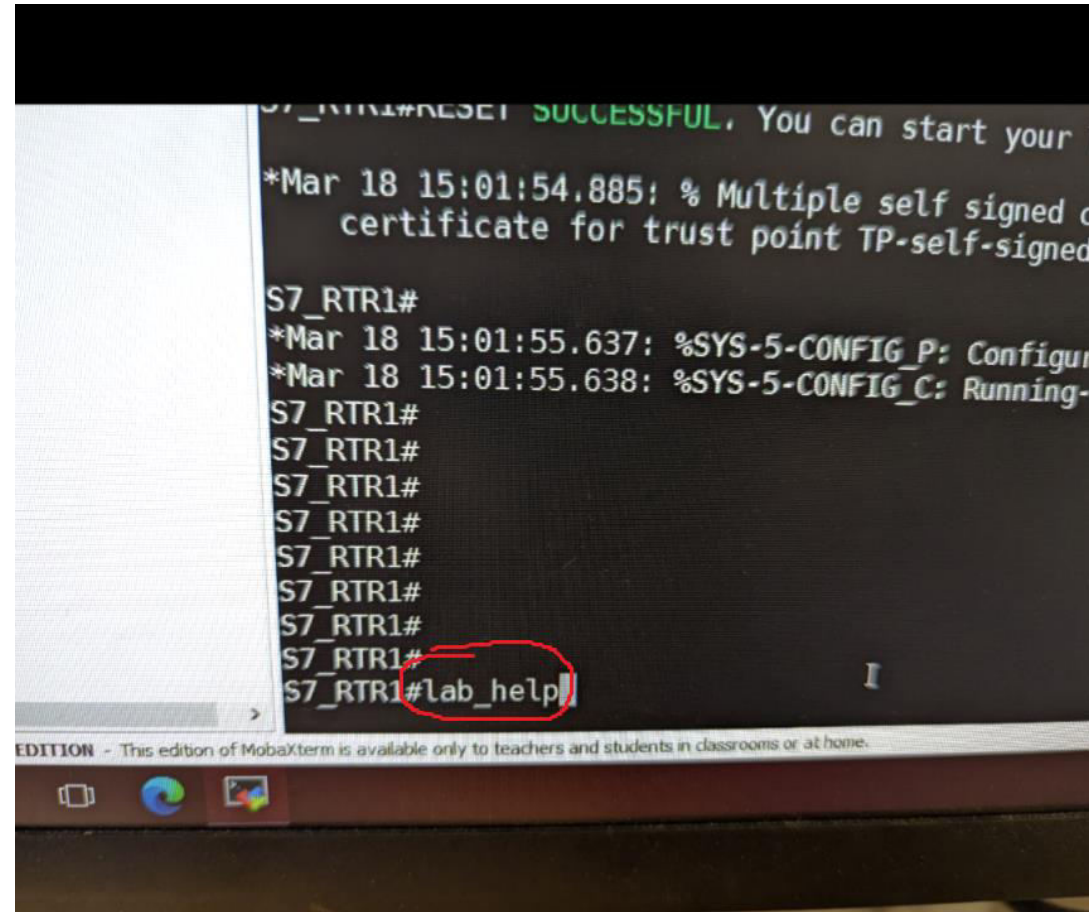
A red arrow points to the line "=====| This is Station 7 -> Router_1 |=====".

Read these instructions

Connecting VMs and Network devices



Connecting VMs and Network devices



The image shows a terminal window with a dark background and white text. The text represents a command-line interface for a Cisco router named S7_RTR1. The session starts with a successful reset message. Then, there are two system messages about self-signed certificates. The user enters the command 'lab_help', which is circled in red. The terminal window has a title bar that says 'EDITION - This edition of MobaXterm is available only to teachers and students in classrooms or at home.' and a taskbar at the bottom with icons for a folder, a web browser, and a file explorer.

```
S7_RTR1#RESET SUCCESSFUL. You can start your l
*Mar 18 15:01:54.885: % Multiple self signed ce
certificate for trust point TP-self-signed-
S7_RTR1#
*Mar 18 15:01:55.637: %SYS-5-CONFIG_P: Configure
*Mar 18 15:01:55.638: %SYS-5-CONFIG_C: Running-c
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#lab_help
```

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Connecting VMs and Network devices

```
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#
S7_RTR1#lab_help

To perform the labs, you will have to
instantiate two VMs with following configuration:

VM_71 -> (192.168.71.71 /24)
VM_72 -> (192.168.72.72 /24)

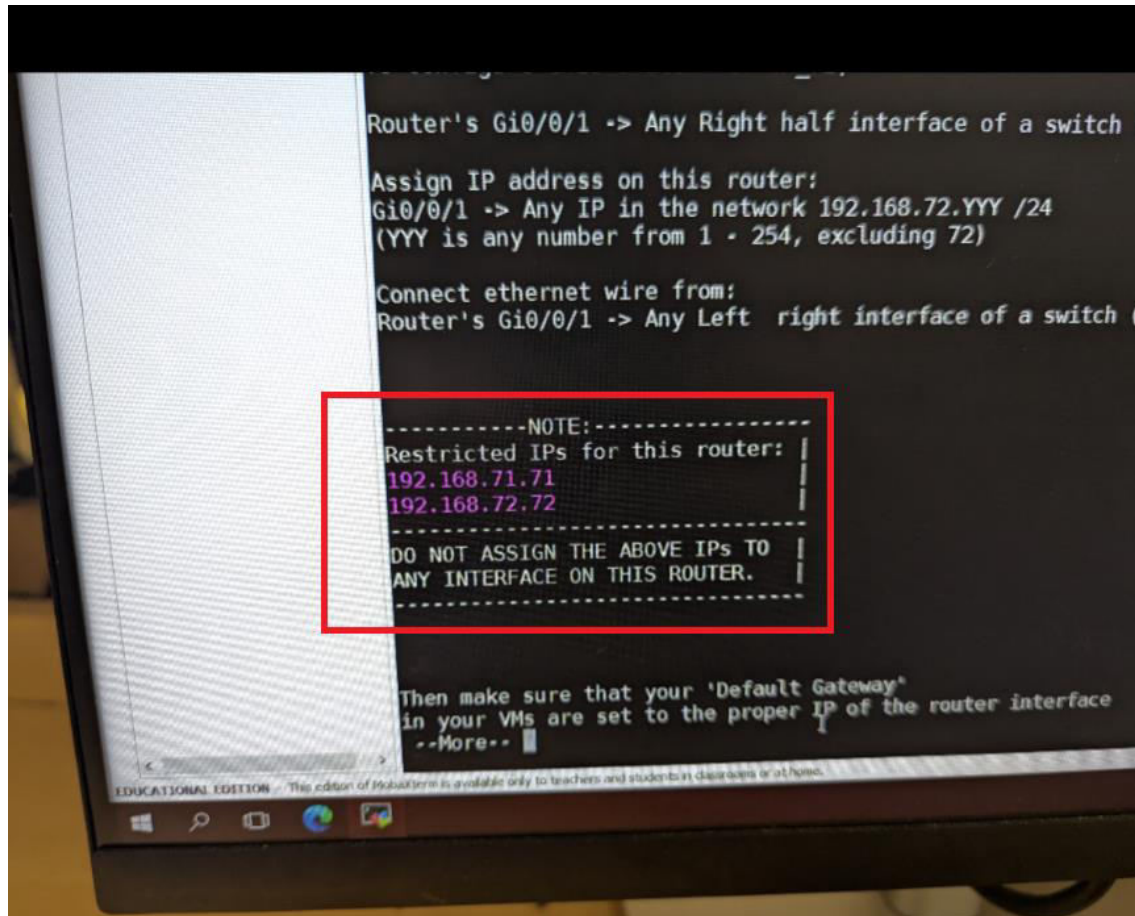
To do this:
1. Press WIN + R.
2. Type 'mstsc' and press 'OK'.
3. Type '10.137.89.71' and press 'Connect'.
4. When login prompt appears, select 'More choices'
5. Select 'Use a different account'
6. Username: cisco
7. Password: cisco
8. when logged in, open 'cmd' and type 'ipconfig /all'
9. Make sure that the IP address of 'Ethernet 3' is -> 192.168.71.71 /24

Perform the above actions again for VM_72
When instantiating, type '10.137.89.72' in step 3.
and Make sure that the IP address of 'Ethernet 3' is -> 192.168.72.72 /24
--More--
```

**READ
THESE
INSTRUCTIONS**

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Connecting VMs and Network devices



Connecting VMs and Network devices

```
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#  
S7_RTR1#show ip interface brief
```

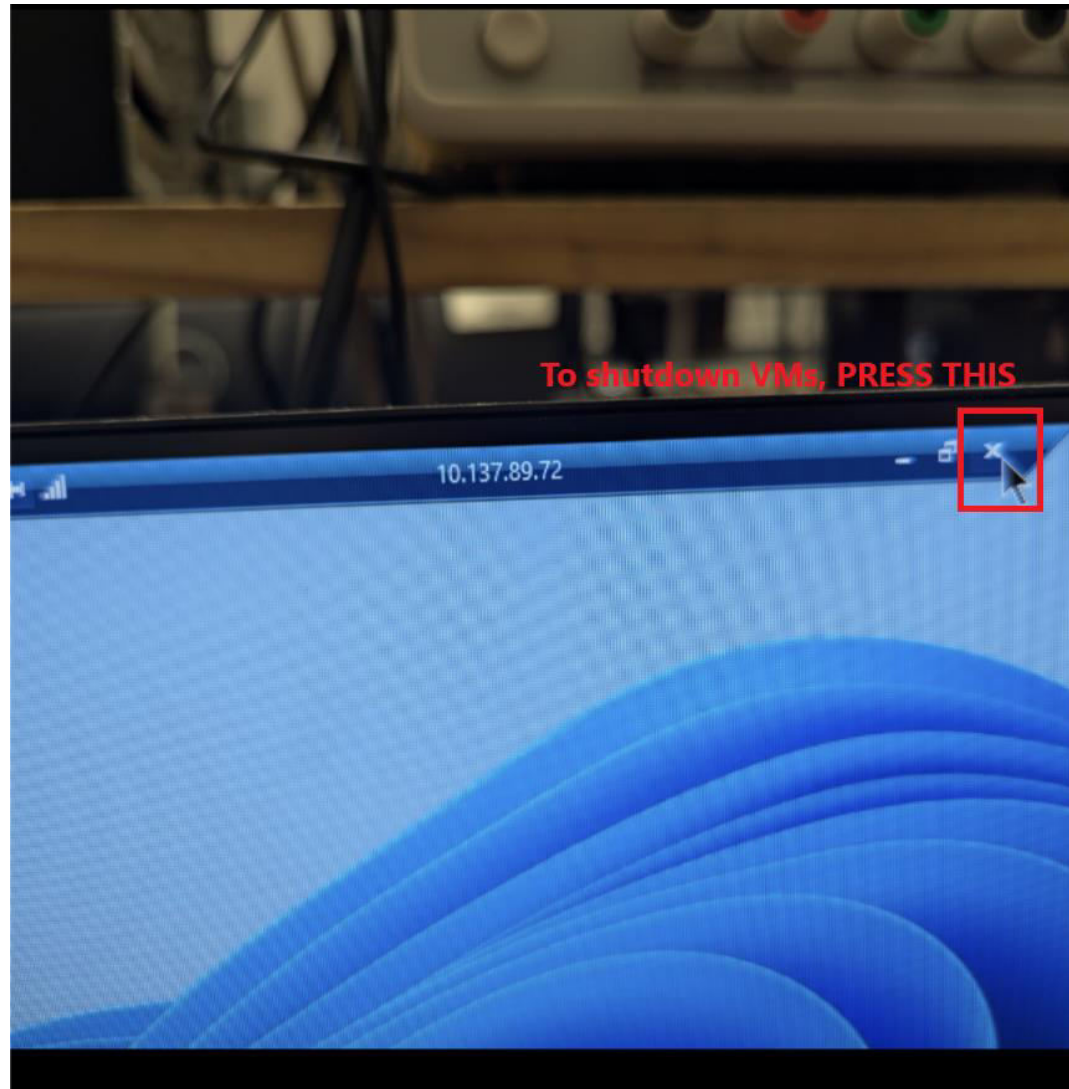
Interface	IP-Address	OK?	Method	Status
GigabitEthernet0/0/0	192.168.71.71	YES	manual	up
GigabitEthernet0/0/1	192.168.72.72	YES	manual	up
GigabitEthernet0/0/2	unassigned	YES	NVRAM	down
GigabitEthernet0/0/3	unassigned	YES	NVRAM	down

```
S7_RTR1#
```

DON'T
DO
THIS!

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Connecting VMs and Network devices



Connecting VMs and Network devices

